# 3.3 Rings and Modules

## 3.3.1 Commutative Rings

**Definition 3.3.1 (Ring)**
*A ring consists of a triple $(A, +, \cdot)$ where $A$ is a set and $+$ and $\cdot$ are laws of composition ("additive" and "multiplicative" respectively) such that the following holds*

- $(A, +)$ *is an* ***abelian group****, whose identity element we refer to as* $0_A$*.*
- $(A, \cdot)$ *is a* ***monoid****, whose identity element we refer to as* $1_A$
- $+$ *and* $\cdot$ *satisfy the* ***distributive property****, that is for all* $x, y, z \in A$

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

$$(x + y) \cdot z = x \cdot z + y \cdot z$$

*For $x \in A$ we write the additive inverse as $-x$, and abbreviate multiplication $x \cdot y =: xy$.*

*We say that $A$ is a* ***zero-ring*** *(or trivial) if $0_A = 1_A \iff A = \{0\}$.*

*$A$ is* ***commutative*** *if in addition $xy = yx$ i.e. $(A, \cdot)$ is abelian.*

**Example 3.3.2**
*The set of integers $\mathbb{Z}$ (see the Integers subsection) is the canonical example of a ring with operations of addition and multiplication.*

**Definition 3.3.3 (Subring)**
*A* ***subring*** *of a ring $A$ is a subset $B$ such that*

- $0_A, 1_A \in B$
- $x \in B \implies -x \in B$
- $x, y \in B \implies x + y \in B$
- $x, y \in B \implies x \cdot y \in B$

*Then $(B, +|_{B\times B}, \cdot\,|_{B\times B})$ is a ring.*

**Definition 3.3.4 (Multiplicative set)**
*A subset $S \subset A$ is said to be* ***multiplicative*** *if*

- $1 \in S$
- $x, y \in S \implies xy \in S$

*Further it is said to be* ***saturated*** *if in addition*

$$x, y \in S \iff xy \in S$$

**Definition 3.3.5 (Integral Domain)**
*A commutative ring $A$ is said to be an* ***integral domain*** *if it is not a zero-ring and it is* cancellative*, that is*

$$ab = ac, a \neq 0 \implies b = c.$$

**Definition 3.3.6 (Reduced)**
*A commutative ring $A$ is said to be* ***reduced*** *if for all $a \in A$*

$$a^n = 0 \implies a = 0$$

**Definition 3.3.7 (Unit / Group of Units)**
*An element $0 \neq a$ of a ring $A$ is called a* ***unit*** *if it has a two-sided multiplicative inverse.*

*For $A$ not a zero-ring, the set of units $A^\star$ forms a group under multiplication, called the* ***group of units****.*

**Definition 3.3.8 (Field)**
*A* ***field*** *$K$ is a commutative non-zero ring such that every non-zero element is a unit, so that $K^\star$ is a group under multiplication and $K^\star = K \setminus \{0\}$.*

**Proposition 3.3.9**
*Every subring of a field $K$ is an integral domain.*

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*Proof.* Suppose $A \subset K$ is a subring. Suppose that $a, b \in A$ such that $ab = 0$. Suppose $a \neq 0$ then $a^{-1}ab = 0 \implies b = 0$. ∎

Note we have the implications

**Corollary 3.3.10**
*Let $A$ be a ring then we have the following implications*

$$\textbf{field} \implies \textbf{integral domain} \implies \textbf{reduced}$$

**Definition 3.3.11 (Ring homomorphism)**
*A **ring homomorphism** $\varphi : A \to B$ is a mapping which is both a multiplicative (monoid) and additive (group) homomorphism*

- $\varphi(0_A) = 0_B$
- $\varphi(1_B) = 1_B$
- $\varphi(x + y) = \varphi(x) + \varphi(y)$
- $\varphi(xy) = \varphi(x)\varphi(y)$

*The **kernel** of $\varphi$ is defined to be*

$$\ker(\varphi) = \{a \mid \varphi(a) = 0_B\}$$

**Definition 3.3.12 (Ideal)**
*A (two-sided) **ideal** $\mathfrak{a}$ of a ring $A$ is a subset of $A$ which is an additive subgroup and closed under multiplication by $A$ :*

- $0_A \in \mathfrak{a}$
- $x, y \in \mathfrak{a} \implies x + y \in \mathfrak{a}$
- $x \in \mathfrak{a} \implies -x \in \mathfrak{a}$
- $x \in \mathfrak{a}, a \in A \implies ax, xa \in \mathfrak{a}$

*$\mathfrak{a}$ is said to be **proper** if $\mathfrak{a} \neq A$.*

**Lemma 3.3.13 (Proper ideal)**
*An ideal $\mathfrak{a}$ is proper if and only if $1 \notin \mathfrak{a}$ if and only if $\mathfrak{a} \cap A^\star = \emptyset$.*

*Alternatively $\mathfrak{a} = A$ if and only if $1 \in \mathfrak{a}$ if and only if $\mathfrak{a} \cap A^\star \neq \emptyset$.*

**Proposition 3.3.14**
*Let $\varphi : A \to B$ be a ring homomorphism, then*

*a) The kernel $\ker(\varphi)$ is a two-sided ideal of $A$*
*b) The image $\varphi(A)$ is a subring of $B$*
*c) $\varphi$ is injective if and only if $\ker(\varphi) = \{0\}$*

**Proposition 3.3.15 (Krull's Theorem)**
*Let $A$ be a ring and $\mathfrak{a}$ a proper ideal. Then it is contained in a proper maximal ideal $\mathfrak{m}$.*

*In particular any non-unit $a \notin A^\star$ is contained in a maximal ideal.*

**Proposition 3.3.16 (Criteria to be a Field)**
*Let $A$ be a ring. Then the following are equivalent*

*a) $A$ is a field*
*b) $A$ is not the zero-ring and the only proper ideal is $\{0\}$.*

*Proof.* a) $\implies$ b). By definition $A$ is not the zero-ring. Let $\mathfrak{a}$ be a proper ideal. By (3.3.13) $\mathfrak{a} \cap A^\star = \emptyset \implies \mathfrak{a} = \{0\}$ as required.

b) $\implies$ a). Suppose $0 \neq a \in A$ then the ideal $Aa = (a)$ is either $\{0\}$ or $A$. However $a = 1 \cdot a \in (a)$ which implies $(a) \neq \{0\}$ and therefore $(a) = A$. In particular there exists $a^{-1} \in A$ such that $a^{-1}a = 1_A$ and $a$ is invertible. ∎

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### 3.3.2 Sums in a Ring

**Proposition 3.3.17 (Distributivity for Sums in a Ring)**
*Let $A$ be a ring, $r \in A$ and $(a_i)_{i\in I}$ a family in $A$ with all but finitely many terms zero. Then $(ra_i)_{i\in I}$ and $(a_i r)_{i\in I}$ again have all but finitely many terms zero, and*

$$r\sum_{i\in I} a_i = \sum_{i\in I} ra_i \qquad \left(\sum_{i\in I} a_i\right) r = \sum_{i\in I} a_i r$$

*If $(b_j)_{j\in J}$ is another such family then $(a_i b_j)_{(i,j)\in I\times J}$ has all but finitely many terms zero, and*

$$\left(\sum_{i\in I} a_i\right)\left(\sum_{j\in J} b_j\right) = \sum_{(i,j)\in I\times J} a_i b_j = \sum_{i\in I}\sum_{j\in J} a_i b_j$$

*and likewise with the order of summation exchanged.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

By the distributive property $x \mapsto rx$ and $x \mapsto xr$ are homomorphisms of the additive group of $A$, so the first claims are (3.2.51) for these homomorphisms. If $a_i b_j \neq 0$ then $a_i \neq 0$ and $b_j \neq 0$, so $a_i b_j$ vanishes outside a product of two finite sets. Write $b := \sum_{j\in J} b_j$. Then by the first part applied to $b$ and then to each $a_i$

$$\left(\sum_{i\in I} a_i\right) b = \sum_{i\in I} a_i b = \sum_{i\in I} a_i \sum_{j\in J} b_j = \sum_{i\in I}\sum_{j\in J} a_i b_j = \sum_{(i,j)\in I\times J} a_i b_j$$

where the last equality is (3.2.53). The other order of summation is identical. ∎

### 3.3.3 Modules I

**Definition 3.3.18 (Module)**
*Let $A$ be a ring. A **left $A$-module** $(M, +, \cdot)$ is an abelian group $(M, +)$ together with a "multiplication" operation*

$$\cdot : A \times M \to M$$

*which satisfies the following properties*

- $(a \times_A a') \cdot x = a \cdot (a' \cdot x)$
- $(a +_A a') \cdot x = a \cdot x + a' \cdot x$
- $a \cdot (x + y) = a \cdot x + a \cdot y$

*Similarly a **right $A$-module** $(M, +, \cdot)$ is an abelian group $(M, +)$ together with a multiplication operation*

$$\cdot : M \times A \to M$$

- $(a \times_A a') \cdot x = (x \cdot a') \times_A a$
- $(a +_A a') \cdot x = a \cdot x + a' \cdot x$
- $a \cdot (x + y) = a \cdot x + a \cdot y$

*Considering the first property $M$ is a left $A$-module iff it is a right $A^{op}$ module in the obvious way. Therefore in the usual case that $A$ is commutative the concepts coincide and we may speak simply of an **$A$-module**, though we almost always write the action on the left.*

*Similarly almost all results for left $A$-modules carry over unchanged for right $A$-modules. In this case we may simply by refer to $A$-modules rather than state the result for both cases separately.*

**Definition 3.3.19 (Submodule)**
*Let $(M, +, \cdot)$ be a left $A$-module. Then a subset $N \subset M$ is called an $A$-**submodule** if*

- *$N$ is a subgroup of $(M, +)$*
- *$m \in N, a \in A \implies am \in N$*

*Then $(N, +|_{N\times N}, \cdot|_{A\times N})$ is a left $A$-module. Similar definition applies for a right $A$-module.*

**Definition 3.3.20 (Module homomorphism)**
*Let $(M, +, \cdot), (N, +, \cdot)$ be left $A$-modules. A function $f : M \to N$ is an $A$-module homomorphism if*

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- *It is an (additive) group homomorphism* $(M,+) \to (N,+)$.
- *It is A-linear;* $\forall a \in A, m \in M \quad f(a \cdot m) = a \cdot f(m)$

*It may be verified that* $f$ *is bijective if and only if it's an isomorphism. In this way we have the following categories*

- $A$-**Mod** *the category of modules over a* commutative *ring* $A$
- ${}_A$**Mod** *the category of left* $A$*-modules*
- **Mod**$_A$ *the category if right* $A$*-modules*

**Definition 3.3.21 (Kernel and Image)**
*The* ***kernel*** *of a module homomorphism* $f$ *is given by*

$$\ker(f) := \{m \in M \mid f(m) = 0\}$$

*and the* ***image*** *is given by*

$$\mathrm{Im}(f) = f(M)$$

**Example 3.3.22 (Trivial Examples)**
*A ring* $A$ *is a left* $A$*-module over itself, denoted* $A_s$.

**Definition 3.3.23 (Restriction of Scalars)**
*Let* $\varphi : A \to B$ *a ring homomorphism and* $M$ *a* $B$*-module. Then we may consider* $M$ *as an* $A$*-module in the obvious way. Denote this by* $[M]_\varphi$.

**Proposition 3.3.24 (Submodules constitute a lattice)**
*Let* $M$ *be an* $A$*-module then the collection* $\mathrm{SubMod}(M)$ *of* $A$*-submodules form a complete sub-lattice of* $\mathcal{P}(M)$ *with meet and join given by*

$$\bigwedge_{i\in I} N_i = \bigcap_{i\in I} N_i$$

*and (the* ***internal sum****)*

$$\bigvee_{i\in I} N_i = \bigcap_{N_i \subseteq N \leq M} N =: \sum_{i\in I} N_i = \left\{ \sum_{j\in J} n_j \;\middle|\; n_j \in N_j \quad \#J < \infty \right\}$$

*Moreover it is the image of the closure operator* $\langle - \rangle : \mathcal{P}(M) \to \mathcal{P}(M)$ *given by*

$$\langle X \rangle = \bigcap_{X \subseteq N} N = \left\{ \sum_j a_j x_j \;\middle|\; x_j \in X \right\}$$

*Proof.* The $A$-submodules of $M$ naturally form a Moore family of subsets of $M$. By (2.1.41) they form a complete sub-lattice with the given form of meet and join. Furthermore it is the image of the given closure operator. The only non-trivial statement is the explicit form of $\sum_{i\in I} N_i$ TODO. ∎

**Lemma 3.3.25**
*Let* $M$ *be a module. Then*

*a)* $\langle \bigcup_{i\in I} X_i \rangle = \sum_{i\in I} \langle X_i \rangle$
*b)* $\langle \bigcup_{i\in I} N_i \rangle = \sum_{i\in I} N_i$
*c)* $N_1 \subseteq N_2 \implies N_1 + N_2 = N_2$

*Proof.*
a) This follows from (2.1.44) applied to the closure operator $\langle - \rangle$
b) This follows from a) because $N_i = \langle N_i \rangle$
c) This follows from b) because $N_1 \cup N_2 = N_2$

∎

**Definition 3.3.26 (Hom Sets)**
*A* ***module homomorphism*** $\varphi : M \to N$ *is an additive group homomorphism which commutes with the* $A$ *action*

$$\varphi(am) = a\varphi(m) \quad \forall a \in A \, m \in M$$

*Denote the abelian group of* $A$*-module homomorphisms*

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$$\mathrm{Hom}_A(M,N)$$

*and the endomorphism ring*

$$\mathrm{End}_A(M) \coloneqq \mathrm{Hom}_A(M,M)$$

*When $A$ is commutative then these have natural $A$-module and $A$-algebra structures respectively.*

### 3.3.4 Operations on Ideals

For this section we assume $A$ is a commutative ring.

**Definition 3.3.27 (Product of ideal and module)**
*Let $M$ be an $A$-module and $\mathfrak{a} \triangleleft A$ an ideal. Define*

$$\mathfrak{a}M = \langle \mathfrak{a} \cdot M \rangle = \left\{ \sum_{i=1}^{n} a_i m_i \;\middle|\; a_i \in \mathfrak{a} \quad m_i \in M \right\}$$

**Proposition 3.3.28 (Lattice of Ideals)**
*Let $A$ be a ring and $\mathcal{I}(A)$ the set of ideals. Then $\mathcal{I}(A)$ forms a complete lattice ordered by inclusion with join and meets given by*

$$\bigwedge_{i\in I} \mathfrak{a}_i = \bigcap_{i\in I} \mathfrak{a}_i$$

*and*

$$\bigvee_{i\in I} \mathfrak{a}_i = \bigcap_{\mathfrak{a}_i \subseteq \mathfrak{a}} \mathfrak{a} =: \sum_i \mathfrak{a}_i \coloneqq \left\{ \sum_i a_i \;\middle|\; a_i \in \mathfrak{a}_i \right\}$$

*This induces a corresponding closure operator*

$$\langle - \rangle : \mathcal{P}(A) \to \mathcal{I}(A)$$

*given by*

$$\langle X \rangle \coloneqq \bigcap_{X \subseteq \mathfrak{a}} \mathfrak{a} = \left\{ \sum_j a_j x_j \;\middle|\; a_j \in A \quad x_j \in X \right\}$$

**Proposition 3.3.29**
*Let $A$ be a ring and $\mathfrak{a}_i$ a family of ideals. Then*

$$\langle \bigcup_{i\in I} \mathfrak{a}_i \rangle = \sum_{i\in I} \mathfrak{a}_i$$

**Definition 3.3.30 (Product of ideals)**
*The product of two ideals $\mathfrak{ab}$ is*

$$\mathfrak{ab} = \left\{ \sum_{i=1}^{n} a_i b_i \;\middle|\; a_i \in \mathfrak{a} \quad b_i \in \mathfrak{b} \right\}$$

*and is itself an ideal.*

**Definition 3.3.31 (Coprime)**
*We say two elements $x, y$ of a commutative ring $A$ are **co-prime** if $(x,y) = (1) \iff ax + by = 1$ for some $a, b \in A$*

*We say a family of ideals $\{\mathfrak{a}_i\}_{i\in I}$ are co-prime if $\sum_{i\in I} \mathfrak{a}_i = A$.*

**Definition 3.3.32 (Principal Ideal)**
*A **principal ideal** is an ideal generated by a single element*

$$(a) \coloneqq \langle \{a\} \rangle = Aa$$

**Lemma 3.3.33**
*A principal ideal $(a)$ is proper if and only if $a \notin A^\star$*

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**Definition 3.3.34 (Maximal Ideal)**
*An ideal $\mathfrak{m} \triangleleft A$ is **maximal** if it is both proper and not contained in another proper ideal.*

**Definition 3.3.35 (Prime Ideal)**
*An ideal $\mathfrak{p} \triangleleft A$ is **prime** if it is both proper and satisfies the following property*

$$xy \in \mathfrak{p} \implies x \in \mathfrak{p} \vee y \in \mathfrak{p}$$

**Definition 3.3.36 (Radical Ideal)**
*An ideal $\mathfrak{a} \triangleleft A$ is **radical** if it satisfies the following property*

$$x^n \in \mathfrak{a} \implies x \in \mathfrak{a}$$

**Proposition 3.3.37 (Maximal ideals exist)**
*Let $A$ be a ring and $\mathfrak{a} \triangleleft A$ a proper ideal. Then it is contained in some maximal ideal $\mathfrak{m}$.*

*In particular there always exists a maximal ideal by considering $\mathfrak{a} = (0)$.*

*Proof.* Simple application of Zorn's Lemma. ∎

**Proposition 3.3.38 (Properties of prime ideals)**
*Let $\mathfrak{p}$ be a prime ideal and $\mathfrak{a}$, $\mathfrak{b}$ be ideals then the following are equivalent*

*a)* $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$
*b)* $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$
*c)* $\mathfrak{ab} \subseteq \mathfrak{p}$

*Proof.* a) $\implies$ b) Follows because $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{a}$

b) $\implies$ c) Follows because $\mathfrak{ab} \subseteq \mathfrak{a} \cap \mathfrak{b}$

c) $\implies$ a) If $\mathfrak{a} \nsubseteq \mathfrak{p}$, then choose $a \in \mathfrak{a} \setminus \mathfrak{p}$. By hypothesis $a\mathfrak{b} \subseteq \mathfrak{p}$ and since $\mathfrak{p}$ is prime $\mathfrak{b} \subseteq \mathfrak{p}$. ∎

**Corollary 3.3.39 (Ideal version of primality)**
*Let $\mathfrak{p}$ be a proper ideal. Then $\mathfrak{p}$ is prime if and only if the following condition holds for all ideals $\mathfrak{a}$, $\mathfrak{b}$*

$$\mathfrak{ab} \subseteq \mathfrak{p} \implies \mathfrak{a} \subseteq \mathfrak{p} \text{ or } \mathfrak{b} \subseteq \mathfrak{p}$$

*In particular for all $k > 0$ we have*

$$\mathfrak{a} \subseteq \mathfrak{p} \iff \mathfrak{a}^k \subseteq \mathfrak{p}$$

*Proof.* One direction has been shown in (3.3.38). Conversely suppose $fg \in \mathfrak{p}$ then apply the condition to the ideals $(f)$ and $(g)$ we find $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. ∎

**Lemma 3.3.40 (Prime ideals are meet-prime)**
*Let $\mathfrak{p}$ be a prime ideal. Then*

$$\bigcap_{i=1}^{n} \mathfrak{a}_i \subseteq \mathfrak{p} \implies \mathfrak{a}_i \subseteq \mathfrak{p} \text{ some } i = 1...n$$

*in other words $\mathfrak{p}$ is meet-prime in the lattice of ideals.*

*Proof.* Suppose $\mathfrak{a}_i \nsubseteq \mathfrak{p}$ for all $i$ then there exists $x_i \in \mathfrak{a}_i \setminus \mathfrak{p}$. Then $x_1...x_n \in \bigcap_{i=1}^{n} \mathfrak{a}_i \subseteq \mathfrak{p}$ by hypothesis, so by primality $x_i \in \mathfrak{p}$ for some $i$, a contradiction. ∎

**Lemma 3.3.41 (Generate prime ideals)**
*Let $A$ be a ring, $S$ a multiplicative set and $\mathfrak{b} \triangleleft A$ such that $\mathfrak{b} \cap S = \emptyset$ then*

$$\mathcal{I} = \{\mathfrak{a} \mid \mathfrak{b} \subseteq \mathfrak{a} \quad \mathfrak{a} \cap S = \emptyset\}$$

*has a maximal element, which is prime.*

*Proof.* Since $\mathfrak{b} \in \mathcal{I}$ it is non-empty. By Zorn's Lemma it has a maximal element, $\mathfrak{p}$. We claim it is prime, for suppose $xy \in \mathfrak{p}$ and $x, y \notin \mathfrak{p}$. Then by maximality $\mathfrak{p} + (x)$ and $\mathfrak{p} + (y)$ intersect $S$. Therefore $S$ intersects $(\mathfrak{p} + (x))(\mathfrak{p} + (y)) \subseteq \mathfrak{p}$, a contradiction. ∎

**Definition 3.3.42 (Minimal prime)**
*Let $A$ be a ring and $\mathfrak{a} \triangleleft A$ a proper ideal. A prime ideal $\mathfrak{p}$ is a **minimal prime over** $\mathfrak{a}$ if it contains $\mathfrak{a}$, and every other such prime ideal contains $\mathfrak{p}$.*

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*We say it is simply a* ***minimal prime*** *if it is minimal over* $(0)$.

**Proposition 3.3.43 (Prime ideals are chain complete)**
*Let* $\{\mathfrak{p}_i\}_{i\in I}$ *be a* ***chain*** *of prime ideals, then* $\bigcap_i \mathfrak{p}_i$ *and* $\bigcup_i \mathfrak{p}_i$ *are prime ideals.*

*Proof.* By (3.3.37) $\bigcup_i \mathfrak{p}_i$ is an ideal, and it's easily verified to be prime. Clearly $\bigcap_i \mathfrak{p}_i$ is an ideal. Suppose $a, b \notin \bigcap_i \mathfrak{p}_i$ then $a \notin \mathfrak{p}_j$ and $b \notin \mathfrak{p}_k$ with $j \leq k$. Then $b \notin \mathfrak{p}_j$, and $ab \notin \mathfrak{p}_j$ by primality, whence $ab \notin \bigcap_i \mathfrak{p}_i$. ■

**Corollary 3.3.44 (Minimal primes exist)**
*Let* $A$ *be a ring and* $\mathfrak{a} \lhd A$ *be a proper ideal contained in a prime ideal* $\mathfrak{p}$*. Then there exists a minimal prime over* $\mathfrak{a}$ *contained in* $\mathfrak{p}$*.*

*In particular there always exists a minimal prime over* $\mathfrak{a}$ *and every prime ideal contains a minimal prime ideal.*

*Proof.* We may use (3.3.43) together with Zorn's Lemma. ■

**Proposition 3.3.45**
*Let* $A$ *be a ring. Then the set* $\mathrm{Rad}(A)$ *of radical ideals forms a complete sub-lattice of the lattice of ideals* $\mathcal{I}(A)$*. This induces a closure operator*

$$\sqrt{-} : \mathcal{I}(A) \to \mathrm{Rad}(A)$$

*given by*

$$\sqrt{\mathfrak{a}} \coloneqq \bigcap_{\mathfrak{a}\subseteq\mathfrak{r}} \mathfrak{r} = \{x \mid x^n \in \mathfrak{a} \quad n > 0\}$$

*The "join" is given by*

$$\bigvee_{i\in I} \mathfrak{a}_i = \sqrt{\sum_i \mathfrak{a}_i}$$

*In particular*

*a)* $\mathfrak{a} \subseteq \sqrt{\mathfrak{a}}$
*b)* $\mathfrak{a} \subseteq \mathfrak{b} \implies \sqrt{\mathfrak{a}} \subseteq \sqrt{\mathfrak{b}}$
*c)* $\sqrt{\sqrt{\mathfrak{a}}} = \sqrt{\mathfrak{a}}$

*Proof.* The set of radical ideals is closed under arbitrary intersections (which are meets in the lattice $\mathcal{I}(A)$). Therefore by (2.1.41) it forms a complete sub-lattice with meet given by intersection of ideals.

It also shows that $\sqrt{-}$ as defined is a closure operator with image $\mathrm{Rad}(A)$, which demonstrates the required properties.

Finally we just need to show that $I' \coloneqq \{x \mid x^n \in \mathfrak{a} \quad n > 0\}$ is equal to $\sqrt{\mathfrak{a}}$. Firstly it's an ideal for if $x, y \in I'$ then $x^n \in \mathfrak{a}$ and $y^m \in \mathfrak{a}$, so we may show that $(x+y)^{n+m} \in \mathfrak{a}$ whence $x + y \in I'$. Similarly $a \in A$ and $x \in I'$ implies $(ax)^n = a^n x^n \in I'$. It's radical for suppose $x^m \in I'$ then $x^{mn} = (x^m)^n \in \mathfrak{a}$ by definition whence $x \in I'$. As it contains $\mathfrak{a}$ we find that $\sqrt{\mathfrak{a}} \subseteq I'$. Let $\mathfrak{r}$ be another radical ideal containing $\mathfrak{a}$ then $x \in I' \implies x^n \in \mathfrak{a} \implies x^n \in \mathfrak{r} \implies x \in \mathfrak{r}$. Therefore the reverse inclusion follows. ■

**Proposition 3.3.46 (Prime Nullstellensatz)**
*The radical of an ideal satisfies*

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{a}\subseteq\mathfrak{p}:\mathfrak{p}\text{ prime}} \mathfrak{p}$$

*Further the intersection may be taken over all* minimal *primes over* $\mathfrak{a}$*.*

*Proof.* Suppose $x \in \sqrt{\mathfrak{a}}$ and $\mathfrak{p} \supseteq \mathfrak{a}$. Then $x^n \in \mathfrak{p} \implies x \in \mathfrak{p}$. Therefore $\sqrt{\mathfrak{a}} \subseteq \bigcap_{\mathfrak{a}\subseteq\mathfrak{p}} \mathfrak{p}$. Conversely suppose $x \notin \sqrt{\mathfrak{a}}$ then $S \coloneqq \{1, x, x^2, \ldots\}$ is a proper multiplicative set such that $S \cap \mathfrak{a} = \emptyset$. By (3.3.41) there is a prime ideal $\mathfrak{p}$ containing $\mathfrak{a}$ which does not intersect $S$. Therefore $x \notin$ RHS as required. ■

**Proposition 3.3.47 (Properties of Radical Ideals)**
*Let* $\mathfrak{a}$, $\mathfrak{b}$ *be ideals then*

*a)* $\sqrt{\mathfrak{a}^k} = \sqrt{\mathfrak{a}}$ *for* $k > 0$
*b)* $\sqrt{\sum_i \mathfrak{a}_i} = \sqrt{\sum_i \sqrt{\mathfrak{a}_i}}$
*c)* $\sqrt{\mathfrak{a}} = A \iff \mathfrak{a} = A$
*d)* $\sum_i \mathfrak{a}_i = A \iff \sum_i \sqrt{\mathfrak{a}_i} = A$
*e)* $\sum_{i=1}^n \mathfrak{a}_i^{k_i} = A \iff \sum_{i=1}^n \mathfrak{a}_i = A \quad k_i > 0.$

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*Proof.*

a) This may be shown by direct calculation or combining (3.3.46) and (3.3.38).
b) This follows by applying (2.1.44) to the closure operator $\sqrt{-}$.
c) $\sqrt{\mathfrak{a}} = A \iff 1 \in \sqrt{\mathfrak{a}} \iff 1 \in \mathfrak{a} \iff \mathfrak{a} = A$
d) This follows from combining c) and b)
e) This follows from d), b) and a)

■

**Lemma 3.3.48 (Ideal Finitely Generated by Nilpotents is Nilpotent)**
*Let $A$ be a ring with ideals $\mathfrak{a}, \mathfrak{b}$ such that $\sqrt{\mathfrak{a}} \subseteq \mathfrak{b}$ and $\mathfrak{b}$ finitely generated. Then there exists an integer $n > 0$ such that $\mathfrak{b}^n \subseteq \mathfrak{a}$.*

**Definition 3.3.49 (Extended and contracted ideals)**
*Let $\varphi : A \to B$ be a homomorphism and $\mathfrak{a}$ (resp. $\mathfrak{b}$) be an ideal of $A$ (resp. $B$). Define the **contraction** (resp. **extension**) ideals as follows*

$$\mathfrak{b}^c := \varphi^{-1}(\mathfrak{b})$$
$$\mathfrak{a}^e := \varphi(\mathfrak{a})B := \langle \varphi(\mathfrak{a}) \rangle = \left\{ \sum_i b_i \varphi(a_i) \;\middle|\; a_i \in \mathfrak{a} \right\}$$

*An ideal is said to be **contracted** (resp. **extended**) if it is of the form $\mathfrak{b}^c$ (resp. $\mathfrak{a}^e$)*

**Proposition 3.3.50 (Operations on ideals)**
*Let $\varphi : A \to B$ a ring homomorphism and $\mathfrak{a} \lhd A$, $\mathfrak{b} \lhd B$ ideals then*

*a) $\mathfrak{b}^c \lhd A$ and $\mathfrak{a}^e \lhd B$*
*b) $\mathfrak{b}^c$ proper if and only if $\mathfrak{b}$ is proper*
*c) $\mathfrak{b}^{ce} \subseteq \mathfrak{b}$ and $\mathfrak{a} \subseteq \mathfrak{a}^{ec}$*
*d) $\mathfrak{a}^{ece} = \mathfrak{a}^e$ and $\mathfrak{b}^{cec} = \mathfrak{b}^c$*
*e) $\mathfrak{b}^{ce} = \mathfrak{b} \iff \mathfrak{b}$ is an extended ideal $\iff \mathfrak{b} \subseteq \mathfrak{b}^{ce}$*
*f) $\mathfrak{a}^{ec} = \mathfrak{a} \iff \mathfrak{a}$ is a contracted ideal $\iff \mathfrak{a}^{ec} \subseteq \mathfrak{a}$*
*g) $\sqrt{\mathfrak{b}^c} = \left(\sqrt{\mathfrak{b}}\right)^c$*
*h) $\left(\sqrt{\mathfrak{b}^c}\right)^e \subseteq \sqrt{\mathfrak{b}}$ with equality when $\varphi$ is surjective*

*When $\varphi$ is surjective every ideal $\mathfrak{b} \lhd B$ is extended, and the contracted ideals are precisely the ideals containing $\ker(\varphi)$.*

*Proof.* We prove each in turn

a)-c) Straightforward

d) By the previous step $\mathfrak{b}^{ce} \subseteq \mathfrak{b} \implies (\mathfrak{b}^{ce})^c \subseteq \mathfrak{b}^c$, similarly $\mathfrak{b}^c \subseteq (\mathfrak{b}^c)^{ec}$. The other relation is similar.

e)-f) These follow from c) and d)

g) $x \in \left(\sqrt{\mathfrak{b}}\right)^c \iff \varphi(x) \in \sqrt{\mathfrak{b}} \iff \varphi(x)^n \in \mathfrak{b} \iff \varphi(x^n) \in \mathfrak{b} \iff x^n \in \mathfrak{b}^c \iff x \in \sqrt{\mathfrak{b}^c}$

h) By c) and g) we find $\left(\sqrt{\mathfrak{b}^c}\right)^e = \left(\sqrt{\mathfrak{b}}\right)^{ce} \subseteq \sqrt{\mathfrak{b}}$. We will show that when $\varphi$ is surjective every ideal is extended, in which case the equality follows from e).

Suppose that $\varphi$ is surjective. Then by e) we only need to show that $\mathfrak{b} \subseteq \mathfrak{b}^{ce}$ for every ideal $\mathfrak{b}$. Let $y \in \mathfrak{b}$ then $y = \varphi(x)$, whence $x \in \mathfrak{b}^c$ and $y \in \mathfrak{b}^{ce}$. ■

**Proposition 3.3.51**
*Let $\varphi : A \to B$ be a ring homomorphism and $\mathfrak{a}_1, ..., \mathfrak{a}_n$ ideals of $A$. Then*

$$\left(\sum_{i=1}^n \mathfrak{a}_i\right)^e = \sum_{i=1}^n \mathfrak{a}_i^e$$

*Similarly*

$$\left(\prod_{i=1}^n \mathfrak{a}_i\right)^e = \prod_{i=1}^n \mathfrak{a}_i^e$$

*Proof.* Evidently $\varphi(\mathfrak{a}_i) \subseteq \varphi\left(\sum_{i=1}^n \mathfrak{a}_i\right) \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e$ and therefore $\mathfrak{a}_i^e \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e \implies \sum_{i=1}^n \mathfrak{a}_i^e \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e$.

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Similarly $\varphi\left(\sum_{i=1}^n \mathfrak{a}_i\right) \subseteq \sum_{i=1}^n \varphi(\mathfrak{a}_i) \subseteq \sum_{i=1}^n \mathfrak{a}_i^e$ whence the reverse inclusion follows.

The second statement follows similarly. ■

**Corollary 3.3.52**
*Let $\varphi : A \to B$ be a ring homomorphism then extension and contraction constitute a monotone Galois connection*

$$\{\mathfrak{a} \lhd A\} \longleftrightarrow \{\mathfrak{b} \lhd B\}$$

*and therefore is order-preserving and satisfies the adjoint property*

$$\mathfrak{a} \subseteq \mathfrak{b}^c \iff \mathfrak{a}^e \subseteq \mathfrak{b}$$

*is satisfied.*

*Proof.* Extension and contraction satisfy conditions of (2.1.50) by (3.3.50) c) and d) ■

**Corollary 3.3.53**
*Let $\varphi : A \to B$ be a ring homomorphism then there is a order-preserving bijection between "contracted" and "extended ideals"*

$$\{\mathfrak{a} \lhd A \mid \mathfrak{a} \text{ contracted}\} \longleftrightarrow \{\mathfrak{b} \lhd B \mid \mathfrak{b} \text{ extended}\}$$

*which restricts to proper ideals.*

*Proof.* We've shown that $\mathfrak{a}$ (resp. $\mathfrak{b}$) is contracted (resp. extended) if and only if the given maps are mutually inverse. Note that $\mathfrak{b}$ is proper implies $\mathfrak{b}^c$ is proper. Furthermore $\mathfrak{b}^c$ proper implies $\mathfrak{b}^{ce} \subseteq \mathfrak{b}$ is proper. Therefore it restricts to proper ideals. ■

**Proposition 3.3.54 (Inverse image of maximal / prime ideals)**
*Let $\varphi : A \to B$ be a morphism then*

- *$\mathfrak{q} \lhd B$ prime $\implies \varphi^{-1}(\mathfrak{q})$ prime*
- *$\mathfrak{n} \lhd B$ maximal and $\varphi$ surjective $\implies \varphi^{-1}(\mathfrak{n})$ is maximal*

**Proposition 3.3.55**
*Consider maps $\varphi : A \to B$ and $\psi : B \to C$ and an ideal $\mathfrak{a} \lhd A$. Then extension of ideals is transitive, that is*

$$\psi(\varphi(\mathfrak{a})B)C = (\psi \circ \varphi)(\mathfrak{a})C$$

## 3.3.5 Quotient Rings

**Proposition 3.3.56 (Quotient Ring)**
*Let $(A, +, \cdot)$ be a ring and $\mathfrak{a}$ an ideal. As $\mathfrak{a}$ is an additive subgroup we may consider the quotient group $(A/\mathfrak{a}, +)$. For an element $a \in A$ write $a + \mathfrak{a}$ for the coset $[a]_\mathfrak{a} \in A/\mathfrak{a}$. There is a well-defined multiplicative law of composition*

$$\cdot : A/\mathfrak{a} \times A/\mathfrak{a} \to A/\mathfrak{a}$$

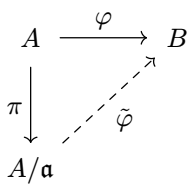
$$(a + \mathfrak{a}) \cdot (b + \mathfrak{a}) \to (a \cdot b + \mathfrak{a})$$

*which makes $(A/\mathfrak{a}, +, \cdot)$ into a ring. Further there is a canonical surjective ring homomorphism*

$$\pi : A \to A/\mathfrak{a}$$

*with the following properties*

- $\ker(\pi) = \mathfrak{a}$
- *Every morphism $\varphi : A \to B$ such that $\mathfrak{a} \subseteq \ker(\varphi)$, factors uniquely through $\pi$.*




- $\ker(\tilde{\varphi}) = \ker(\varphi)/\mathfrak{a}$
- *$\tilde{\varphi}$ is injective if and only if* $\ker(\varphi) = \mathfrak{a}$
- *$\tilde{\varphi}$ is surjective if and only if $\varphi$ is surjective*

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*For an ideal $\mathfrak{b} \supseteq \mathfrak{a}$ define the corresponding quotient ideal*

$$\mathfrak{b}/\mathfrak{a} := \{b + \mathfrak{a} \mid b \in \mathfrak{b}\} = \pi(\mathfrak{b})$$

*This induces a bijective, order-preserving correspondence of ideals*

$$\{\mathfrak{b}' \lhd A/\mathfrak{a}\} \underset{\pi(-)}{\overset{\pi^{-1}(-)}{\rightleftarrows}} \{\mathfrak{b} \lhd A \mid \mathfrak{a} \subseteq \mathfrak{b}\}$$

*under which maximal (resp. prime, radical) ideals of $A$ containing $\mathfrak{a}$ correspond to maximal (resp. prime, radical) ideals of $A/\mathfrak{a}$.*

**Corollary 3.3.57 (Isomorphism Theorem)**
*Let $\varphi : A \to B$ be a ring homomorphism. Then this induces a canonical isomorphism*

$$A/\ker(\varphi) \cong \varphi(A) \subset B$$

**Corollary 3.3.58 (Second Isomorphism Theorem)**
*Let $\mathfrak{b}$,$\mathfrak{a}$ be ideals then there is a unique morphism making the diagram commute*

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ {\scriptstyle\pi}\downarrow & \swarrow & \downarrow{\scriptstyle\pi} \\ A/(\mathfrak{a}+\mathfrak{b}) & \xrightarrow{\sim} & (A/\mathfrak{a})/((\mathfrak{a}+\mathfrak{b})/\mathfrak{a}) \end{array}$$

*which is in fact an isomorphism. If $\mathfrak{a} + \mathfrak{b} \subseteq \mathfrak{c}$ this restricts to an isomorphism of $A/\mathfrak{b}$-modules*

$$\begin{array}{ccc} \mathfrak{c} & \xrightarrow{\pi} & \mathfrak{c}/\mathfrak{a} \\ {\scriptstyle\pi}\downarrow & \swarrow & \downarrow{\scriptstyle\pi} \\ \mathfrak{c}/(\mathfrak{a}+\mathfrak{b}) & \xrightarrow{\sim} & (\mathfrak{c}/\mathfrak{a})/((\mathfrak{a}+\mathfrak{b})/\mathfrak{a}) \end{array}$$

**Proposition 3.3.59 (Criteria for Maximal, Prime and Reduced)**
*Let $\mathfrak{a} \lhd A$ then $\mathfrak{a}$ is*

- ***maximal** if and only if $A/\mathfrak{a}$ is a field*
- ***prime** if and only if $A/\mathfrak{a}$ is an integral domain*
- ***radical** if and only if $A/\mathfrak{a}$ is reduced*

*Proof.* Suppose $\mathfrak{a}$ is maximal then it is by definition proper so $A/\mathfrak{a}$ is not the zero-ring. By (3.3.56) then $A/\mathfrak{a}$ has no proper non-zero ideals and so by (3.3.16) is a field.

Conversely if $A/\mathfrak{a}$ is a field it is by definition not the zero ring, and so by (3.3.56) $\mathfrak{a}$ is proper. Furthermore by the same result $\mathfrak{a}$ is maximal.

Suppose $\mathfrak{a}$ is prime and $\overline{x} \cdot \overline{y} = 0$. Then by definition $\overline{x \cdot y} = 0 \implies x \cdot y \in \mathfrak{a} \implies x \in \mathfrak{a}$ or $y \in \mathfrak{a} \implies \overline{x} = 0$ or $\overline{y} = 0$. This shows that $A/\mathfrak{a}$ is an integral domain. The converse is similar. ∎

**Corollary 3.3.60**
*Let $\mathfrak{a} \lhd A$ be a proper ideal, then the following implications hold*

$$\textbf{maximal} \implies \textbf{prime} \implies \textbf{radical}$$

*Proof.* This follows by combining (3.3.10) and (3.3.59). ∎

**Corollary 3.3.61 (Field Morphisms are injective)**
*Let $\varphi : k \to B$ be a homomorphism from a field to a non-zero ring. Then $\varphi$ is injective.*

*Proof.* $\ker(\varphi)$ is an ideal. As $\varphi(1_k) = 1_B$ and $0_B \neq 1_B$ then $\ker(\varphi) \neq k$. Since the only ideals are $(0)$ and $k$ we see $\ker(\varphi) = \{0\}$ and $\varphi$ is injective. ∎

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### 3.3.6 Chinese Remainder Theorem

**Proposition 3.3.62 (Chinese Remainder Theorem)**
*Let $A$ be a commutative ring with ideals $\mathfrak{a}_1, ..., \mathfrak{a}_n$ which are pairwise coprime (i.e. $\mathfrak{a}_i + \mathfrak{a}_j = A$ for all $1 \leq i < j \leq n$). Then there is a canonical isomorphism*

$$A/\bigcap_{i=1}^{n} \mathfrak{a}_i \longrightarrow A/\mathfrak{a}_1 \times ... \times A/\mathfrak{a}_n$$

*Further we have the following identity*

$$\bigcap_{i=1}^{n} \mathfrak{a}_i = \prod_{i=1}^{n} \mathfrak{a}_i$$

*Proof.* Consider the canonical ring homomorphism

$$\varphi : A \longrightarrow A/\mathfrak{a}_1 \times ... \times A/\mathfrak{a}_n.$$

Trivially it has kernel $\bigcap_{i=1}^{n} \mathfrak{a}_i$. Therefore by (3.3.56) it is sufficient to show it is surjective. First we show that $e_1 := (1, 0, ..., 0)$ is in the image of $\varphi$. For we have $x_i + y_i = 1$ for $x_i \in \mathfrak{a}_1$, $y_i \in \mathfrak{a}_i$ and $1 < i \leq n$. Take $f_1 = y_2 ... y_n \in \bigcap_{i=2}^{n} \mathfrak{a}_i$; since $y_i = 1 - x_i$, $f_1 = \prod_{i=2}^{n}(1 - x_i) \in 1 + \mathfrak{a}_1$ (as each $x_i \in \mathfrak{a}_1$), so combined with $f_1 \in \bigcap_{i=2}^{n} \mathfrak{a}_i$, $\varphi(f_1) = e_1$ as required.

Similarly there exists $f_i$ such that $\varphi(f_i) = (0, ..., 1, ..., 0) =: e_i$. Finally for $\overline{x_1} \in A/\mathfrak{a}_1, ..., \overline{x_n} \in A/\mathfrak{a}_n$ then

$$\varphi(f_1 x_1 + ... + f_n x_n) = (\overline{x_1}, ..., \overline{x_n})$$

For the final statement we may assume by induction that

$$\bigcap_{i=1}^{n-1} \mathfrak{a}_i = \prod_{i=1}^{n-1} \mathfrak{a}_i$$

We have equations

$$x_i + y_i = 1 \quad x_i \in \mathfrak{a}_i, y_i \in \mathfrak{a}_n$$

Then

$$x_1 ... x_{n-1} = \prod_{i=1}^{n-1}(1 - y_i) \in 1 + \mathfrak{a}_n$$

so we may conclude

$$\bigcap_{i=1}^{n-1} \mathfrak{a}_i + \mathfrak{a}_n = A.$$

This then reduces to the case $n = 2$. Suppose $x \in \mathfrak{a} \cap \mathfrak{b}$, by assumption $1 = y + z$ for $y \in \mathfrak{a}$ and $z \in \mathfrak{b}$. Then $x = x(y + z) \in \mathfrak{ab}$, so $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{ab}$. The reverse inclusion is obvious. ∎

**Proposition 3.3.63 (Chinese Remainder Theorem II)**
*Let $A$ be a commutative ring with $\mathfrak{a}_1, ..., \mathfrak{a}_n$ co-prime ideals. For any positive integers $k_1, ..., k_n$ there is a canonical isomorphism*

$$A/\bigcap_{i=1}^{n} \mathfrak{a}_i^{k_i} \longrightarrow A/\mathfrak{a}_1^{k_1} \times ... \times A/\mathfrak{a}_n^{k_n}$$

*and*

$$\bigcap_{i=1}^{n} \mathfrak{a}_i^{k_i} = \prod_{i=1}^{n} \mathfrak{a}_i^{k_i}$$

*Proof.* By (3.3.47) the ideals $\mathfrak{a}_1^{k_1}, ..., \mathfrak{a}_n^{k_n}$ are pairwise co-prime. Therefore we may apply (3.3.62). ∎

### 3.3.7 Irreducible and Reduced rings

We say an element $x$ is nilpotent if $x^n = 0$. By (3.3.45) these form an ideal.

**Definition 3.3.64 (Nilradical)**
*Define the **nilradical** to be the set (ideal) of nilpotents*

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$$N(A) := \sqrt{(0)} \overset{(3.3.46)}{=} \bigcap_{\mathfrak{p}} \mathfrak{p}$$

*Clearly $A$ is reduced if and only if $N(A) = \{0\}$.*

We also make the following definition

**Definition 3.3.65 (Irreducible)**
*Let $A$ be a ring. We say $A$ is **irreducible** if $N(A)$ is prime.*

The notion of irreducible ring is related to the notion of minimal primes

**Proposition 3.3.66**
*A ring $A$ is irreducible if and only if it has a unique minimal prime ideal. In this case it is equal to $N(A)$.*

*Proof.* First we note that every prime ideal contains $N(A)$. If $A$ is irreducible then by definition $N(A)$ is prime and it is therefore the unique minimal prime ideal.

Conversely if $\mathfrak{p}_0$ is the unique minimal prime ideal then by (3.3.44) it is contained in every prime ideal. Therefore (3.3.64) $N(A) = \bigcap_{\mathfrak{p}} \mathfrak{p} = \mathfrak{p}_0$ is prime and $A$ is irreducible. ∎

**Proposition 3.3.67 (Integral Domain $\iff$ Reduced and Irreducible)**
*Let $A$ be a ring. Then the following are equivalent*

*a) $A$ is an integral domain*
*b) $A$ is reduced and has a unique minimal prime*
*c) $(0)$ is prime*
*d) $A$ is reduced and irreducible.*

The following may be useful

**Proposition 3.3.68**
*Let $A$ be a ring then the sum of an invertible and nilpotent element is again invertible*

$$A^\star + N(A) \subseteq A^\star$$

*Proof.* For if

$$a = u + n = u(1 - (u^{-1})(-n))$$

with $n \in N(A)$ and $u \in A^\star$. Then way may reduce to the case $1 - n$ and observe that

$$(1-n)^{-1} = \sum_{i=0}^{\infty} n^i$$

which by assumption is a finite sum. ∎

## 3.3.8 Algebra over a Commutative Ring

For what follows let $A$ be a commutative ring.

**Definition 3.3.69 (Algebra (over a commutative ring))**
*An algebra over $A$ (or an $A$-algebra) is a pair $(i_B, B)$ where $B$ is a (not necessarily commutative) ring and $i_B : A \to B$ is a ring homomorphism.*

*We call $i_B$ the structural morphism and write $a \cdot b := i_B(a)b$*

*Morphisms of $A$-algebras are the ring homomorphisms $\varphi : B \to C$ such that $\varphi \circ i_B = i_C$. This then constitutes a category $\mathbf{Alg}_A$.*

*If $k$ is a field an algebra over $k$ is referred to as a $k$-algebra.*

**Definition 3.3.70 (Sub-algebra)**
*Let $(i_B, B)$ be an $A$-algebra. A sub-algebra $C$ is a subring $C$ of $B$ for which*

$$a \in A \quad c \in C \implies i_B(a)c \in C$$

**Example 3.3.71 (Algebra over commutatve sub-ring)**
*If $A \subset B$ is a commutative sub-ring, then $B$ is naturally a $A$-algebra.*

*The polynomial ring $A[X]$ is naturally an $A$-algebra*

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**Definition 3.3.72 (Algebra generated by a set)**
*Let $B$ be an $A$-algebra. The collection of $A$-subalgebras forms a Moore family. Therefore by (2.1.41) there is a canonical closure operator*

$$A[-] : \mathcal{P}(B) \to \mathrm{SubAlg}_A(B)$$

*which we denote by $A[S]$ for $S \subset B$. A more explicit characterization when $S$ is finite is given in the Polynomial Rings in Many Variables subsection. More generally we have*

$$A[S] = \bigcup_{S' \subset S | S' \text{ finite}} A[S']$$

**Proposition 3.3.73**
*Let $B = A[S]$ be an $A$-algebra and $\mathfrak{a}$ a sub-$A$-module of $B$. Then $\mathfrak{a}$ is an ideal if and only if*

$$s \in S \implies s\mathfrak{a} \subseteq \mathfrak{a}$$

*Proof.* One direction is obvious. Suppose the condition given holds, and define

$$B' := \{b \in B \mid b\mathfrak{a} \subseteq \mathfrak{a}\}$$

Then clearly $S \subseteq B'$. It's easy to show that $B'$ is a sub-$A$-algebra of $B$, so $B' = B$ and $\mathfrak{a}$ is an ideal. ∎

## 3.3.9 Finite-type Algebras

**Definition 3.3.74 (Finite algebra)**
*An $A$-algebra $B$ is* finite *if it is finite as an $A$-module.*

**Definition 3.3.75 (Finitely generated algebra)**
*An $A$-algebra $B$ is finitely generated (or of finite type) if there exists an integer $n \in \mathbb{N}$ and a surjection of $A$-algebras*

$$A[X_1, ..., X_n] \to B$$

*the images of $X_i$ are the generators.*

**Proposition 3.3.76**
*Let $C$ be a finitely-generated $B$ algebra and $B$ a finitely-generated $A$-algebra. Then $C$ is finitely-generated as an $A$-algebra.*

*Proof.* Suppose $C = B[c_1, ..., c_n]$. Then by definition $c_i = F_i(b_1, ..., b_m)$ for some $F_i \in A[X_1, ..., X_m]$. Then we may take as a generating set

$$b_1^{\lambda_1} ... b_m^{\lambda_m}$$

where $\lambda \in \mathbb{N}^m$ runs through all the exponents of non-zero coefficients of the $F_i$. ∎

## 3.3.10 Bimodules

For applications it is often useful to have multiple rings acting on a single module ("bimodule"). For example an $A$-algebra $B$ naturally has an action from both $A$ and $B$. It may also allow us to generalise results which would otherwise only hold in the commutative case.

**Definition 3.3.77 (Bimodule)**
*Let $A, B$ be rings. We say an abelian group $M$ is a $(A, B)$-**bimodule** if it is both a left $A$-module and a right $B$-module for which the two actions commute*

$$a \cdot (m \cdot b) = (a \cdot m) \cdot b \quad \forall a \in A, b \in B, m \in M$$

*We may denote this by ${}_AM_B$ in order to emphasise the actions. If homomorphisms are defined in the obvious way then these constitute a category ${}_A\mathbf{Mod}_B$. Recall that every abelian group is automatically both a left and right $\mathbb{Z}$-module with the following action for $N \in \mathbb{Z}$*

$$N \cdot m = m \cdot N := \begin{cases} \underbrace{m + ... + m}_{N \text{ times}} & N \geq 0 \\ -(m \cdot (-N)) & N < 0 \end{cases}$$

*Therefore we have the following generalisations*

- *A right $B$-module is precisely a $(\mathbb{Z}, B)$-bimodule*

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- *A left $A$-module is precisely a $(A,\mathbb{Z})$-bimodule*
- *When $A$ is commutative an $A$-module has a well-defined $(A,A)$-bimodule structure by defining*

$$a\cdot m\cdot a' := a'a\cdot m = aa'\cdot m$$

  *in other words $A$-**Mod** is (equivalent to) a full subcategory of ${}_A\mathbf{Mod}_A$.*
- *A ring $A$ has an obvious $(A,A)$-bimodule structure by associativity of the multiplication operation*

*We will understand by the notation $N_B$ that $N$ is a $(\mathbb{Z},B)$-bimodule and by ${}\{\}_A M$ that $M$ is a $(A,\mathbb{Z})$-bimodule.*

We generalize slightly the consideration of the third bullet point

**Proposition 3.3.78 ($(A,A)$-Bimodule $\equiv$ $A$-module)**
*Let $\varphi : A\to B$ be a homomorphism of commutative rings and $Z$ a $B$-module. Then $Z$ is naturally a $(B,A)$-module with the following action*

$$b\cdot m\cdot a := \varphi(a)b\cdot m \qquad (\star)\quad \forall a\in A, b\in B, m\in Z$$

*Suppose $M$ is another $(B,A)$-bimodule satisfying $(\star)$ then we may identify the $(B,A)$-bimodule homomorphisms and the left $B$-module homomorphisms.*

$$\mathrm{Hom}(M,\,{}_BZ_A)\xrightarrow{\sim}\mathrm{Hom}({}_BM,\,{}_BZ)$$

*In particular when $\varphi = 1_A$ and $B=A$ we may identify $(A,A)$-bimodules and $A$-modules, as well as the corresponding homomorphisms. Explicitly for $M,Z$ $A$-modules these have well-defined $(A,A)$-bimodule structures and there is a bijection*

$$\mathrm{Hom}({}_AM_A,\,{}_AZ_A)\xrightarrow{\sim}\mathrm{Hom}(M,Z)$$

*Proof.* We may show that the right $A$-module structure on $Z$ is well-defined because $B$ is commutative and using the associativity of the $B$-module action. Similarly the associativity of the $B$-module action ensures that the actions commute and so form a $(B,A)$-bimodule structure.

Suppose $\theta : M\to Z$ is a (left) $B$-module homomorphism. Then

$$\theta(m\cdot a)=\theta(\varphi(a)\cdot m)=\varphi(a)\theta(m)=\theta(m)\cdot a$$

so it is $(B,A)$-bilinear as required. The converse is clear, so we see that the sets are equal as subsets of the set of abelian group homomorphisms.

The case $\varphi = 1_A$ is immediate. ∎

We generalize the notion of "Hom-Set"

**Definition 3.3.79 (Hom Functors)**
*Let $A,B,C$ be arbitrary rings. Then we have an enriched hom functor for ${}_A\mathbf{Mod}_B$*

$$\mathrm{Hom} : {}_A\mathbf{Mod}_B^{op}\times{}_A\mathbf{Mod}_B\to\mathbf{AbGrp}$$

*In order to generalise the Tensor-Hom adjunction we consider the following functors ("right-linear" and "left-linear" respectively)*

$$\mathrm{RHom} : {}_B\mathbf{Mod}_A^{op}\times{}_C\mathbf{Mod}_A\to{}_C\mathbf{Mod}_B$$
$$\mathrm{LHom} : {}_A\mathbf{Mod}_B^{op}\times{}_A\mathbf{Mod}_C\to{}_B\mathbf{Mod}_C$$

*For example if $\psi\in\mathrm{RHom}({}_BM_C,\,{}_AN_C)$ then the $(A,B)$-bimodule action is defined to be*

$$(a\psi b)(m) := a\psi(bm)$$

*and similarly if $\varphi\in\mathrm{LHom}({}_AM_B,\,{}_AN_C)$ then*

$$(b\varphi c)(m) := \varphi(mb)c$$

*which we may verify satisfies the axioms of a bimodule. In order to standardize notation we may write $\mathrm{Hom}_A$ in place of* RHom *and* LHom.

*If $B=C=\mathbb{Z}$ then we recover the simple case (3.3.26).*

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*If $A$ is a commutative ring then we've observed that $A$-**Mod** is a full subcategory of ${}_A\mathbf{Mod}_A$ and the functors* RHom, LHom *become equal to the simple case (3.3.26) when restricted to this subcategory.*

### 3.3.11 Module Direct Product and Sum

**Definition 3.3.80 (External Direct Product / Sum)**
*Let $A$ be a ring and $\{M_i\}_{i\in I}$ a family of $A$-modules. Define the **external direct product** as the set of ordered tuples indexed over $I$*

$$\prod_{i\in I} M_i \coloneqq \left\{ (m_i)_{i\in I} \;\middle|\; m_i \in M_i \right\}$$

*with the obvious module operations. The **external direct sum** is the subset of tuples for which all but finitely many elements are zero. We denote this as follows*

$$\bigoplus_{i\in I} M_i$$

*Clearly when $I$ is finite then these are equal.*

**Proposition 3.3.81 (Categorical Product)**
*Let $\{M_i\}_{i\in I}$ be a family of left $A$-modules and consider the family of projections*

$$\pi_i : \prod_{i\in I} M_i \to M_i$$

*For $Z$ an $(A,C)$-bimodule there is a natural isomorphism of left $C$-modules*

$$\begin{aligned} \mathrm{Hom}_A\left(Z, \prod_{i\in I} M_i\right) &\xrightarrow{\sim} \prod_{i\in I} \mathrm{Hom}_A(Z, M_i) \\ \theta &\to (\pi_i \circ \theta)_{i\in I} \end{aligned}$$

*This in particular includes the case $A = C$ is commutative.*

**Proposition 3.3.82 (Categorical Coproduct)**
*Let $\{M_i\}_{i\in I}$ be a family of left $A$-modules and consider the family of inclusions*

$$u_i : M_i \to \bigoplus_{i\in I} M_i$$

*For $Z$ a $(A,C)$-bimodule there is a natural isomorphism of right $C$-modules*

$$\begin{aligned} \mathrm{Hom}_A\left(\bigoplus_{i\in I} M_i, Z\right) &\xrightarrow{\sim} \prod_{i\in I} \mathrm{Hom}_A(M_i, Z) \\ \theta &\to (\theta \circ u_i)_{i\in I} \end{aligned}$$

*This in particular includes the case $A = C$ is commutative.*

**Corollary 3.3.83**
*Let $(M_i)_{i\in I}$ and $(N_j)_{j\in J}$ be families of left $A$-modules. Then there is an isomorphism of abelian groups*

$$\begin{aligned} \mathrm{Hom}_A\left(\bigoplus_{i\in I} M_i, \prod_{j\in J} N_j\right) &\cong \prod_{(i,j)\in I\times J} \mathrm{Hom}_A(M_i, N_j) \\ \psi &\to (\pi_j \circ \psi \circ u_i)_{(i,j)} \end{aligned}$$

*where $\pi_j : \prod_{j\in J} N_j \to N_j$ and $u_i : M_i \to \bigoplus_{i\in I} M_i$ are the canonical projections and injections respectively.*

*These maps are $A$-linear if $A$ is commutative.*

**Definition 3.3.84 (Free Module)**
*An $A$-module $M$ is*

- ***free** if it is isomorphic to*

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$$\bigoplus_{i\in A} A =: A^{(I)}$$

*for some indexing set $I$*

- ***finite free*** *if it is free with respect to a* finite *indexing set $I$*

*Under the isomorphism $M \to \bigoplus_{i\in I} A$ the set of elements $\{m_i\}_{i\in I}$ corresponding to the standard basis vectors $e_i$ is called a* ***basis*** *for $M$.*

*We say that $M$ has* ***rank*** *$n$ if there exists a basis of order $n$. NB rank is not necessarily unique.*

**Proposition 3.3.85**
*Let $M$ be an $(A, C)$-bimodule then there is a canonical isomorphism of right $C$-modules*

$$\begin{aligned}\mathrm{Hom}_A(A, M) \cong \quad & M \\ \theta \to \quad & \theta(1_A)\end{aligned}$$

*This in particular includes the case $A = C$ is commutative.*

**Proposition 3.3.86**
*Let $M$ be a free left $A$-module with basis $(m_i)_{i\in I}$ and $N$ an $(A, C)$-bimodule then there is an isomorphism of right $C$-modules*

$$\begin{aligned}\mathrm{Hom}_A(M, N) \cong \quad & \prod_{i\in I} N \\ \theta \to \quad & (\theta(m_i))_{i\in I}\end{aligned}$$

*This in particular includes the case $A = C$ is commutative.*

## 3.3.12 Free Modules

**Definition 3.3.87 (Faithful Module)**
*We say an $A$-module $M$ is* ***faithful*** *if*

$$am = 0 \quad \forall m \in M \implies a = 0$$

**Definition 3.3.88 (Linearly Independent, Spanning and Basis)**
*Let $M$ be an $A$-module and $S \subset M$ a set. We say $S$ is*

- ***spanning*** *if $\langle S\rangle = M$*
- ***linearly independent*** *if for every finite subset $\{s_1, ..., s_n\} \subseteq S$ with $s_i$ distinct we have*

$$\sum_{i=1}^{n} a_i s_i = 0 \implies a_i = 0 \quad 1 \le i \le n$$

- *a* ***basis*** *if it is both spanning and linearly independent*

**Definition 3.3.89 (Finite Module)**
*An $A$-module $M$ is* ***finite*** *if there exists a finite spanning set.*

**Definition 3.3.90 (Minimal spanning set)**
*Let $M$ be an $A$-module. Then $S \subset M$ is a* ***minimal spanning set*** *if it generates $M$ and no proper subset does so.*

**Definition 3.3.91 (Free Module)**
*Let $M$ be an $A$-module. We say that $M$ is a* ***free module*** *over $A$ if it has a basis.*

**Proposition 3.3.92 (Free $A$-module is an external sum of $A$)**
*An $A$-module $M$ is free if and only if it is isomorphic to $\bigoplus_{i\in I} A$ for some $I$. The isomorphism is given by*

$$\sum_{i\in I} a_i m_i \to (a_i)_{i\in I}$$

**Definition 3.3.93 (Internal Direct Sum Module)**
*An $A$-module $M$ is an* ***internal direct sum*** *of submodules $\{M_i\}_{i\in I}$ the canonical mapping of the external direct sum*

$$\bigoplus_{i\in I} M_i \to M$$

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*is an isomorphism.*

**Proposition 3.3.94**
*Let $M$ be an $A$-module and $\{M_i\}_{i\in I}$ a family of submodules. Then the following are equivalent*

*a) $\sum_{i\in I} M_i$ is the internal direct sum of the family $\{M_i\}_{i\in I}$*
*b) The relation $\sum_{i\in I} m_i = 0$ implies $m_i = 0$ for all $i \in I$*
*c) For any $i \in I$ we have $M_i \cap \left(\sum_{k\neq i} M_k\right) = \{0\}$*

**Proposition 3.3.95**
*We say that $M_1, M_2$ are **supplementary** submodules of $M$ if $M$ is the internal direct sum of $M_1$ and $M_2$.*

*We say that $M_1$ is a **direct factor** of $M$ if it is supplementary to another submodule.*

## 3.3.13 Exact Sequences

**Definition 3.3.96 (Vector space)**
*If $k$ is a field and $V$ a $k$-module, then we say $V$ is a **vector space** over $k$.*

**Remark 3.3.97**
*We will see that every vector space is free and every $k$-submodule is a direct factor.*

**Proposition 3.3.98 (Kernel)**
*Let $\varphi : M \to N$ be an $A$-module homomorphism, then the **kernel** of $\varphi$*

$$\ker(\varphi) \coloneqq \{m \in M \mid \varphi(m) = 0\}$$

*is an $A$-submodule of $M$. Observe $\varphi$ is injective iff $\ker(\varphi) = 0$.*

**Proposition 3.3.99 (Image)**
*Let $\varphi : M \to N$ be an $A$-module homomorphism then the image*

$$\mathrm{Im}(\varphi) = \{\varphi(m) \mid m \in M\}$$

*is an $A$-submodule of $N$.*

**Definition 3.3.100 (Quotient Module)**
*Let $N \subseteq M$ be an $A$-submodule then define the **quotient module** $M/N$ to be the quotient group with an action of $A$ given by*

$$a(m + N) = (am + N)$$

*When $N \subseteq P \subseteq M$ is a sequence of submodules then define the $A$-submodule $P/N$ of $M/N$ by*

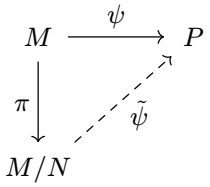
$$P/N \coloneqq \{p + N \mid p \in P\}$$

**Proposition 3.3.101 (Quotient Module Properties)**
*Let $N \subseteq M$ be an $A$-submodule then there is a canonical surjective morphism*

$$\pi : M \to M/N$$

*with the following properties*

*a) $\pi(m) = m + N$*
*b) $\ker(\pi) = N$*
*c) Every homomorphism $\psi : M \to P$ such that $N \subseteq \ker(\psi)$, factors uniquely through $\pi$*




*Furthermore there is a bijection of $A$-submodules*

$$\{P' \subseteq M/N\} \leftrightarrow \{P \mid N \subseteq P \subseteq M\}$$

*given by $P' = P/N$. In the situation above $\ker(\tilde{\psi}) = \ker(\psi)/N$. In particular if $\ker(\psi) = N$ then $\tilde{\psi}$ is injective.*

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**Corollary 3.3.102**
*Let $\psi : M \to N$ be an $A$-module homomorphism, then this induces an isomorphism*

$$M/\ker(\psi) \cong \mathrm{Im}(\psi)$$

**Definition 3.3.103 (Exact Sequence)**
*Let $N \xrightarrow{\varphi} M \xrightarrow{\psi} P$ be an sequence of $A$-module homomorphisms. We say it is **exact** if*

$$\mathrm{Im}(\varphi) = \ker(\psi)$$

*It is equivalent to the following two conditions*

*a)* $\psi \circ \varphi = 0$
*b)* $\psi(m) = 0 \implies m = \varphi(n)$ *for some* $n \in N$.

*An exact sequence of the form*

$$0 \to N \to M \to P \to 0$$

*is said to be **short-exact**.*

**Remark 3.3.104**
*There are a few trivial observations*

- $0 \to M \to N$ *is exact if and only if the map* $M \to N$ *is injective*
- $M \to N \to 0$ *is exact if and only if the map* $M \to N$ *is surjective.*

**Proposition 3.3.105 (Isomorphism induced by short-exact sequence)**
*Let $N \subseteq M$ be a $A$-submodule then there is a canonical short-exact sequence*

$$0 \to N \to M \to M/N \to 0$$

*Conversely suppose we have a short exact sequence*

$$0 \to N \xrightarrow{i} M \xrightarrow{\pi} P \to 0$$

*then this induces an isomorphism*

$$M/i(N) \cong P$$

*If $N$ is a submodule of $M$ then we would simply write $M/N \cong P$.*

**Proposition 3.3.106 (Third Isomorphism Theorem)**
*Let $N \subseteq N' \subseteq M$ be a chain of modules then there is a short-exact sequence*

$$0 \to N'/N \to M/N \to M/N' \to 0$$

*which then induces an isomorphism*

$$(M/N)/(N'/N) \cong M/N'$$

**Proposition 3.3.107 (Product of ideal and quotient module)**
*Let $N$ be a submodule of $M$ and $\mathfrak{a} \lhd A$ an ideal. Then*

$$\mathfrak{a}(M/N) = (N + \mathfrak{a}M)/N$$

**Proposition 3.3.108 (Induced module)**
*Let $M$ be an $A$-module and $\mathfrak{a}$ an ideal such that $\mathfrak{a}M = 0$, then $M$ is naturally an $A/\mathfrak{a}$-module with action given by*

$$\overline{a} \cdot m := a \cdot m$$

### 3.3.14 Dual Module

**Proposition 3.3.109 (Dual Module)**
*Let $M$ be a left (resp. right) $A$-module. Then the set*

$$M^\vee := \mathrm{Hom}_A(M, A)$$

*is canonically a* right *(resp.* left*) $A$-module.*

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*If $M$ is a finite free left (resp. right) $A$-module with basis $\{v_1, ..., v_n\}$ then $M^\vee$ is a finite free* right *(resp.* left*) $A$-module with basis $\{v_1^\vee, ..., v_n^\vee\}$ where these are the unique homomorphisms satisfying*

$$v_i^\vee(v_j) = \delta_{ij}$$

*Moreover every basis of $M^\vee$ is of this form.*

**Proposition 3.3.110 (Hom-Set is free)**
*Let $M$ be a finite free left $A$-module with basis $\{v_1, ..., v_n\}$ and $N$ a $(A, B)$-bimodule. Then there is an isomorphism of right $B$-modules*

$$\begin{aligned} \mathrm{Hom}_A(M, N) \cong & \bigoplus_{i=1}^{n} N \\ \theta \to & (\theta(v_i))_{i \in I} \end{aligned}$$

*If $A$ is commutative and $N$ is a finite-free $A$-module with basis $\{w_1, ..., w_m\}$ then there is further an isomorphism of $A$-modules*

$$\begin{aligned} \mathrm{Hom}_A(M, N) \cong & \bigoplus_{i,j=1}^{n,m} A \\ \theta \to & \left(w_j^\vee(\theta(v_i))\right)_{i,j} \end{aligned}$$

*where $w_1^\vee, ..., w_m^\vee$ is the dual basis for $N^\vee$. In particular $\mathrm{Hom}_A(M, N)$ is a finite-free $A$-module with basis*

$$\left\{w_j v_i^\vee\right\}_{i,j}$$

**Definition 3.3.111 (Dual Functor)**
*Let $A$ be a commutative ring and $\varphi : M \to N$ an $A$-module homomorphism. Define the dual homomorphism $\varphi^\vee : N^\vee \to M^\vee$ by*

$$\varphi^\vee(\psi) := \psi \circ \varphi$$

*This determines a contravariant functor*

$$(-)^\vee : A\textbf{-Mod} \to A\textbf{-Mod}$$

**Corollary 3.3.112 (Double Dual Natural Isomorphism)**
*Let $A$ be a commutative ring and $M$ a finite free $A$-module then the canonical $A$-module homomorphism*

$$\begin{aligned} \eta : M \longrightarrow & \quad M^{\vee\vee} \\ x \to & \quad (\varphi \to \varphi(x)) \end{aligned}$$

*is an isomorphism, which is natural in $M$.*

**Corollary 3.3.113**
*The contravariant functor $(-)^\vee : \mathbf{FiniteFreeMod}_A \to \mathbf{FiniteFreeMod}_A$ is an equivalence of categories and therefore full and faithful.*

*Proof.* Use the dual isomorphism $\eta$ together with (2.6.29) and (2.6.28). ∎

## 3.3.15 Matrices

For this section we assume that $A$ is a commutative ring. In this context $A^n := A \times ... \times A$ is a finite free module with basis $e_1, ..., e_n$. Matrices are concrete realisations of linear maps of finite free modules.

**Proposition 3.3.114**
*Let $M, N$ be finite free $A$-modules with ordered bases $\mathcal{B} := \{v_1, ..., v_n\}$, $\mathcal{B}' := \{w_1, ..., w_m\}$ respectively. Let $\varphi : M \to N$ be an $A$-linear map and define $[\varphi]$ to be the unique $m \times n$ matrix satisfying the following condition*

$$\varphi(v_i) = \sum_{j=1}^{m} [\varphi]_{ji} w_j \quad \forall i = 1...n$$

*(**notice sum over the first index**). Then it is also the unique matrix ensuring that the following diagram commutes*

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$$\begin{array}{ccc} M & \xrightarrow{\varphi} & N \\ \uparrow \sim & & \uparrow \sim \\ A^n & \xrightarrow{x \to [\varphi]x} & A^m \end{array}$$

*where the bottom arrow is given by usual multiplication of a matrix by a column vector*

**Proposition 3.3.115 (Matrices as linear maps)**
*Let $M$, $N$ be free $A$-modules with ordered bases $\mathcal{B} := \{v_1, ..., v_n\}$, $\mathcal{B}' := \{w_1, ..., w_m\}$ respectively. Then there are mutually inverse isomorphisms of $A$-modules*

$$\begin{aligned} \mathrm{Mat}_{m\times n}(A) &\longleftrightarrow \mathrm{Hom}_A(M, N) \\ E &\to \hat{E} \\ [\varphi]^{\mathcal{B}}_{\mathcal{B}'} &\leftarrow \varphi \end{aligned}$$

*where*

$$\begin{aligned} \hat{E}\left(\sum_{i=1}^{n} \lambda_i v_i\right) &:= \sum_{j=1}^{m}\left(\sum_{i=1}^{n} E_{ji}\lambda_i\right) w_j \\ \varphi(v_i) &= \sum_{j=1}^{m} [\varphi]_{ji} w_j \end{aligned}$$

*If we further consider a free $A$-module $P$ with ordered bases $\mathcal{B}'' = \{u_1, ..., u_p\}$ then*

$$\begin{aligned} \hat{E} \circ \hat{F} &= \widehat{EF} \\ [\psi \circ \varphi]^{\mathcal{B}}_{\mathcal{B}''} &= [\psi]^{\mathcal{B}'}_{\mathcal{B}''}[\varphi]^{\mathcal{B}}_{\mathcal{B}'} \end{aligned}$$

*Observe that*

$$[1_M]^{\mathcal{B}}_{\mathcal{B}} = I_n$$

*Furthermore there is an isomorphism of $A$-algebras*

$$\begin{aligned} \mathrm{End}_A(M) &\longleftrightarrow \mathrm{Mat}_{n,n}(A) \\ \varphi &\to \left(v_i^\vee(\varphi(v_j))\right)_{ij} \\ \sum_{ij} E_{ij} v_i v_j^\vee &\leftarrow E \end{aligned}$$

**Corollary 3.3.116**
*Matrix multiplication is associative. In particular*

$$(EF)v = E(Fv)$$

*Proof.* We may consider the free $A$-modules $A^n$, $A^m$ and $A^p$ with canonical bases. The result follows because function composition is associative and $\hat{\cdot}$ is injective. ∎

**Corollary 3.3.117**
*There is an isomorphism of $A$-modules*

$$\begin{aligned} \mathrm{Mat}_{m\times n}(A) &\longleftrightarrow \mathrm{Hom}_A(A^n, A^m) \\ E &\to (v \to Ev) \end{aligned}$$

*and further an isomorphism of $A$-algebras*

$$\mathrm{Mat}_{n,n}(A) \longleftrightarrow \mathrm{End}_A(A^n)$$

**Corollary 3.3.118**
*Let $M$ be a finite free $A$-module with bases $\mathcal{B}, \mathcal{B}'$ and $\varphi \in \mathrm{End}_A(M)$. Then $\varphi$ is an isomorphism if and only if $[\varphi]^{\mathcal{B}}_{\mathcal{B}'} \in \mathrm{Mat}_{n\times n}(A)$ is invertible.*

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**Corollary 3.3.119 (Change of basis)**
*Let $M$ be a finite free $A$-module and $\mathcal{B}, \mathcal{B}'$ bases then*

$$[1_M]^{\mathcal{B}}_{\mathcal{B}'} = \left([1_M]^{\mathcal{B}'}_{\mathcal{B}}\right)^{-1}$$

*and*

$$[\varphi]^{\mathcal{B}'}_{\mathcal{B}'} = P[\varphi]^{\mathcal{B}}_{\mathcal{B}}P^{-1}$$

*where*

$$P := [1_M]^{\mathcal{B}}_{\mathcal{B}'}$$

*is invertible.*

**Definition 3.3.120 (Transpose)**
*Let $E$ be an $m \times n$ matrix in $A$, then define the **transpose** of $E$ to be the $n \times m$ matrix $E^t$ where*

$$\left(E^t\right)_{ij} := E_{ji}$$

**Proposition 3.3.121**
*Let $M, N$ be finite-free $A$-modules with bases $\mathcal{B} = \{v_1, ..., v_n\}$ and $\mathcal{B}' = \{w_1, ..., w_m\}$. Let $\varphi : M \to N$ be an $A$-module homomorphism and $\varphi^\vee : N^\vee \to M^\vee$ the dual homomorphism then*

$$[\varphi^\vee]^{\mathcal{B}'^\vee}_{\mathcal{B}^\vee} = \left([\varphi]^{\mathcal{B}}_{\mathcal{B}'}\right)^t$$

*Similarly if $E$ is an $m \times n$ matrix over $A$ then*

$$\hat{E}^\vee = \widehat{E^t}$$

*where the right hand side is understood to be with respect to the dual bases.*

**Corollary 3.3.122**
*Let $E, F$ be matrices then*

$$(EF)^t = (FE)^t$$

### 3.3.16 Vector Spaces

**Definition 3.3.123 (Vector Space)**
*A **vector space** $V$ over $k$ is simply a $k$-module.*

*A $k$-submodule is referred to as a **subspace***

*A $k$-module homomorphism is referred to as a **linear map***

*A vector space is finite-dimensional if it is finite as a $k$-module.*

The main result on vector spaces is that bases exist and all have the same cardinality. Recall that $\langle\cdot\rangle$ is a closure operator. We show that $(V, \langle\cdot\rangle)$ determines a matroid so that we may appeal to results in the Matroids section. First we need to show that the notions of independence coincide

**Proposition 3.3.124 (Equivalent definitions of linear independence)**
*Let $V$ be a vector space and $S \subset V$. Then the following are equivalent*

*a) $S$ is linearly independent*
*b) No proper subset $S' \subset S$ satisfies $\langle S' \rangle = \langle S \rangle$*
*c) **Matroid Independence** $x \in S \implies x \notin \langle S \setminus \{x\} \rangle$*

*Further $S$ is independent if and only if every finite subset of $S$ is.*

*Proof.* b) $\iff$ c) This is b).

a) $\implies$ c). If $x \in \langle S \setminus \{x\} \rangle$ then it's clear that $S$ is not linearly independent.

c) $\implies$ a). Suppose we have a linear relationship

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$$0 = \sum_{i=1}^{n} \lambda_i v_i \quad v_i \in S$$

By renumbering assume that $\lambda_1 \neq 0$, then rearrange to show $v_1 \in \langle S \setminus \{v_1\} \rangle$, contradicting the hypothesis. ∎

We show that $(V, \langle \cdot \rangle)$ satisfies the Exchange Property and therefore constitutes a matroid.

**Proposition 3.3.125 (Exchange Property)**
*Let $V$ be a vector space and $S \subset V$. Then*

$$y \in \langle S \cup \{x\} \rangle \setminus \langle S \rangle \implies x \in \langle S \cup \{y\} \rangle$$

*Proof.* Suppose $y$ is as given, then

$$y = \lambda x + \sum_i \lambda_i s_i \quad s_i \in S$$

The case $\lambda = 0$ is obvious. Otherwise we may rearrange to find

$$x = \lambda^{-1} y - \sum_i \lambda^{-1} \lambda_i s_i$$

whence $x \in \langle S \cup \{y\} \rangle$. ∎

Therefore we have the following

**Proposition 3.3.126 (Vector Spaces are Free)**
*Every vector space has a basis, and in the finite-dimensional case every basis is finite of the same size. We denote this by $\dim_k V$.*

*More generally every linearly independent set is contained in a basis (so has order at most $\dim_k V$) and every spanning set contains a basis (so has order at least $\dim_k V$)*

*Proof.* Follows from (2.3.7) and (2.3.12). The final statement follows from (2.3.13). ∎

**Proposition 3.3.127 (Basis Criteria)**
*Let $V$ be a vector space with $n = \dim_k V$ and $\mathcal{B} \subseteq V$. Then TFAE*

*a) $\mathcal{B}$ is a basis*
*b) $\mathcal{B}$ is linearly independent and $\#\mathcal{B} \geq \dim_k V$*
*c) $\mathcal{B}$ is spanning and $\#\mathcal{B} \leq \dim_k V$*

*Proof.* The equivalence of a), b) and c) follows from (2.3.14). ∎

**Proposition 3.3.128**
*A vector space $V = \{0\}$ if and only if $\dim_k V = 0$*

**Proposition 3.3.129 (Image of a basis)**
*Let $\varphi : V \to W$ be a linear map*

*a) If $S$ is linearly-independent and $\varphi$ is injective, then $\varphi(S)$ is linearly-independent*
*b) If $\Gamma$ is spanning then ($\varphi$ is surjective $\iff$ $\varphi(\Gamma)$ is spanning)*
*c) If $\mathcal{B}$ is a basis then ($\varphi$ is an isomorphism $\iff$ $\varphi(\mathcal{B})$ is a basis and $\varphi$ injective on $\mathcal{B}$)*

*Proof.*

a) Suppose $\sum_i \lambda_i \varphi(s_i) = 0 \implies \varphi\left(\sum_i \lambda_i s_i\right) = 0$. As $\varphi$ is injective this implies $\sum_i \lambda_i s_i = 0 \implies \lambda_i = 0$.
b) If $\varphi$ is surjective then for $w \in W$ we have $\varphi(v) = w$ for some $v \in V$. By hypothesis $v = \sum_i \lambda_i \gamma_i$ and $w = \sum_i \varphi(\lambda_i)$. Conversely given $w \in W$ by hypothesis $w = \sum_i \lambda_i \varphi(\gamma_i) = \varphi\left(\sum_i \lambda_i \gamma_i\right)$ and $\varphi$ is surjective as required.
c) Suppose $\varphi$ is isomorphism, then it's surely injective on $\mathcal{B}$ and by a),b) $\varphi(\mathcal{B})$ is a basis. Conversely if $\varphi(\mathcal{B}) =: \mathcal{B}'$ is a basis then by b) $\varphi$ is surjective. Suppose $\varphi(v) = 0$. Then by hypothesis $v = \sum_i \lambda_i v_i$ for $v_i \in \mathcal{B}$ and $0 = \varphi(v) = \sum_i \lambda_i \varphi(v_i)$. By hypothesis $\varphi(v_i)$ are distinct elements of the basis $\mathcal{B}'$ and therefore $\lambda_i = 0$ and $v = 0$. Therefore $\varphi$ is injective and hence bijective.

∎

**Corollary 3.3.130 (Dimension is an invariant)**
*Dimension is preserved under isomorphism. More generally for $\varphi : V \to W$ we have*

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$$\varphi \text{ injective} \implies \dim_k V \leq \dim_k W$$

$$\varphi \text{ surjective} \implies \dim_k V \geq \dim_k W$$

**Proposition 3.3.131**
*Let $W \subseteq V$ be finite-dimensional vector spaces then the dimension of the quotient module satisfies*

$$\dim_k V/W = \dim_k V - \dim_k W$$

*Proof.* Let $\{v_1, ..., v_m\}$ be a basis of $W$, then there exists a basis $\{v_1, ..., v_m, v_{m+1}, ..., v_n\}$ of $V$ containing the first by (3.3.126). We claim that

$$\{[v_{m+1}], ..., [v_n]\}$$

is a basis for $V/W$, and the result follows. For given $[v] \in V/W$ then

$$v = \sum_{i=1}^{n} \lambda_i v_i$$

since the basis is spanning. We have

$$v - \sum_{i=m+1}^{n} \lambda_i v_i \in W$$

therefore

$$[v] = \left[\sum_{i=m+1}^{n} \lambda_i v_i\right] = \sum_{i=m+1}^{n} \lambda_i [v_i]$$

and the given set is spanning. Similarly suppose

$$\sum_{i=m+1}^{n} \lambda_i [v_i] = 0$$

then by definition $\sum_{i=m+1}^{n} \lambda_i v_i \in W$. Therefore

$$\sum_{i=m+1}^{n} \lambda_i v_i = \sum_{i=1}^{m} \lambda_i v_i$$

and since $v_i$ are linearly independent we must have $\lambda_i = 0$. ∎

**Proposition 3.3.132 (Injective Criteria)**
*Let $\varphi : V \to W$ be a linear map then*

$$\varphi \text{ injective} \iff \ker(\varphi) = \{0\} \iff \dim_k \ker(\varphi) = 0$$

*Proof.* Note for any linear map $\varphi$ we have $\varphi(0) = 0$. Therefore $\varphi$ injective clearly shows $\ker(\varphi) = \{0\}$. Conversely suppose $\ker(\varphi) = 0$ and $\varphi(v) = \varphi(w)$. Then $\varphi(v - w) = 0 \implies v - w = 0 \implies v = w$ as required. ∎

**Definition 3.3.133 (Rank)**
*Let $\varphi : V \to W$ be a linear map then define*

$$\operatorname{rank}_k(\varphi) := \dim_k(\operatorname{Im}(\varphi))$$

**Proposition 3.3.134 (Surjective Criteria)**
*Let $\varphi : V \to W$ be a linear map with $W$ finite-dimensional then*

$$\varphi \text{ surjective} \iff \operatorname{rank}_k(\varphi) = \dim_k W$$

*Proof.* This follows directly from (2.3.15). ∎

**Proposition 3.3.135 (Isomorphism Theorem / Rank-Nullity)**
*Let $\varphi : V \to W$ be a linear map then this induces an isomorphism*

$$V/\ker(\varphi) \xrightarrow{\sim} \operatorname{Im}(\varphi)$$

*in particular when $V$ is finite-dimensional*

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$$\dim_k V = \dim_k \ker(\varphi) + \operatorname{rank}_k(\varphi)$$

**Corollary 3.3.136 (Isomorphism Criteria)**
*Let $V, W$ vector spaces with $W$ finite-dimensional. A linear map $\varphi : V \to W$ is an isomorphism if and only if any* two *of the following are satisfied*

*a)* $\dim_k \ker(\varphi) = 0 \iff \varphi$ *injective*
*b)* $\dim_k V = \dim_k W$
*c)* $\operatorname{rank}_k(\varphi) = \dim_k W \iff \varphi$ *surjective*

*Proof.* The rank-nullity equation ensures that if any two hold the third is automatically satisfied. In this case $\varphi$ is isomorphism as required. ∎

**Corollary 3.3.137 (Endomorphism Isomorphism Criteria)**
*Let $V$ be a finite-dimensional vector space and $\varphi : V \to V$ then TFAE*

*a)* $\varphi$ *is injective*
*b)* $\varphi$ *is surjective*
*c)* $\varphi$ *is an isomorphism*

*Proof.* For the equivalence of a), b) and c) we may use the previous result with $W = V$ and note $\dim_k W = \dim_k V$ is automatically satisfied. ∎

**Proposition 3.3.138 (Internal Direct Sum)**
*Let $U_1, U_2$ be two subspaces of $V$ then TFAE*

*a)* $U_1 \cap U_2 = \{0\}$ *and* $V = U_1 + U_2$
*b) Every* $v \in V$ *may be written uniquely as* $u_1 + u_2$ *for* $u_i \in U_i$.

*and we say $V = U_1 \oplus U_2$ is an* ***internal direct sum*** *and $U_2$ is a* ***supplementary subspace*** *for $U_1$.*

**Proposition 3.3.139 (Subspaces are direct factors)**
*Every subspace $U$ has a supplementary subspace $U'$ such that*

$$V = U \oplus U'$$

*Proof.* Let $\mathcal{B}_1$ be a basis for $U$ and extend to a basis $\mathcal{B}$ and define $\mathcal{B}_2 := \mathcal{B} \setminus \mathcal{B}_1$. Then it's easy to show that $U' = \langle \mathcal{B}_2 \rangle$ is a supplementary subspace. ∎

**Proposition 3.3.140 (Dimension formula for direct sums)**
*Suppose $V = U_1 \oplus U_2$, $\mathcal{B}_1$ is a basis for $U_1$ and $\mathcal{B}_2$ is a basis for $U_2$. Then $\mathcal{B}_1 \cap \mathcal{B}_2 = \emptyset$ and $\mathcal{B}_1 \cup \mathcal{B}_2$ is a basis for $V$. In particular*

$$\dim_k V = \dim_k U_1 + \dim_k U_2$$

#### 3.3.16.1 Dual Space

**Definition 3.3.141 (Dual Space)**
*Let $V$ be a $k$-vector space and define the* ***dual space*** *to be*

$$V^\vee := \operatorname{Hom}_k(V, k)$$

*This is an abelian group and even a $k$-vector space under the obvious operations. The construction $V \to V^\vee$ determines a contravariant functor*

$$(-)^\vee : \mathbf{Vect}_k \to \mathbf{Vect}_k$$

**Definition 3.3.142 (Annihilator)**
*Let $V$ be a vector space and $U \subseteq V$ a subspace. Define the* ***annihilator*** *of $U$ by*

$$U^\circ = \{\theta \in V^\vee \mid \theta(u) = 0 \quad \forall u \in U\}$$

*This is a linear subspace of $V^\vee$.*

**Proposition 3.3.143 (Dimension formula for annihilators)**
*There is a canonical isomorphism by restriction*

$$V^\vee / U^\circ \to U^\vee$$

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*In particular when $V$ is a finite-dimensional vector space then*

$$\dim_k V = \dim_k U + \dim_k U^\circ$$

*Proof.* Let $W$ be a supplementary subspace and consider the morphism $V = U \oplus W \xrightarrow{\pi_U} U$. Then $(\theta \circ \pi_U)|_U = \theta$ so the restriction map is surjective. Clearly the kernel is $U^\circ$. The dimension formula follows from (3.3.135) and (3.3.109). ∎

**Corollary 3.3.144 (Dual rank = rank)**
*Let $\varphi : V \to W$ be a linear map and $\varphi^\vee : W^\vee \to V^\vee$ then*

$$\begin{aligned} \ker(\varphi^\vee) &= \operatorname{Im}(\varphi)^\circ \\ \operatorname{Im}(\varphi^\vee) &\subseteq \ker(\varphi)^\circ \end{aligned}$$

*In the finite-dimensional case $\operatorname{Im}(\varphi^\vee) = \ker(\varphi)^\circ$ and*

$$\begin{aligned} \dim_k \ker(\varphi^\vee) &= \dim_k W - \operatorname{rank}_k(\varphi) \\ \operatorname{rank}_k(\varphi^\vee) &= \operatorname{rank}_k(\varphi) \end{aligned}$$

*Proof.* Note $\ker(\varphi^\vee) = \operatorname{Im}(\varphi)^\circ$ and $\operatorname{Im}(\varphi^\vee) \subseteq \ker(\varphi)^\circ$ by the definitions.

Consider the finite-dimensional case. By (3.3.143)

$$\dim_k \ker(\varphi^\vee) = \dim_k \operatorname{Im}(\varphi)^\circ = \dim_k W - \operatorname{rank}_k(\varphi)$$

By rank-nullity applied to $\varphi^\vee$ and $\dim_k W = \dim_k W^\vee$ we deduce

$$\operatorname{rank}_k(\varphi^\vee) = \operatorname{rank}_k(\varphi).$$

By (3.3.143) and rank-nullity applied to $\varphi$

$$\dim_k \ker(\varphi)^\circ = \dim_k V - \dim_k \ker(\varphi) = \operatorname{rank}_k(\varphi).$$

Finally by (2.3.15) $\operatorname{Im}(\varphi^\vee) = \ker(\varphi)^\circ$. ∎

From this it follows that taking duals reflects and preserve isomorphisms

**Corollary 3.3.145 ($(-)^\vee$ reflects isomorphisms)**
*Let $\varphi : V \to W$ be a linear map of finite-dimensional spaces then*

*a) $\varphi$ is injective if and only if $\varphi^\vee$ is surjective*
*b) $\varphi$ is surjective if and only if $\varphi^\vee$ is injective*
*c) $\varphi$ is iso if and only if $\varphi^\vee$ is iso*

*Proof.* Note by (3.3.144) we have $\operatorname{rank}_k(\varphi) = \operatorname{rank}_k(\varphi^\vee)$ and by (3.3.109) $\dim_k V^\vee = \dim_k V$.

$\varphi$ is surjective $\iff \operatorname{rank}_k(\varphi) = \dim_k W = \operatorname{rank}_k(\varphi^\vee) \overset{(3.3.135)}{\iff} \dim_k \ker(\varphi^\vee) = 0 \iff \ker(\varphi^\vee) = \{0\}$

$\varphi$ is injective $\iff \dim_k \ker(\varphi) = 0 \overset{(3.3.135)}{\iff} \operatorname{rank}_k(\varphi) = \dim_k V \iff \dim_k V^\vee = \operatorname{rank}_k(\varphi^\vee) \iff \varphi^\vee$ is surjective.

The last point may be deduced from the first two, or the fact that $(-)^\vee$ is full and faithful (3.3.113) and category-theoretic result (2.6.42). ∎

### 3.3.16.2 Bilinear Pairings

**Definition 3.3.146 (Bilinear maps)**
*Let $V, W$ be vector spaces a bilinear map $\psi$ is a map*

$$\psi : V \times W \to k$$

*which is $k$-linear in each variable. We denote the set of bilinear maps as*

$$\operatorname{Bilin}_k(V, W)$$

**Proposition 3.3.147 (Matrix Representation)**
*Let $V, W$ be finite-dimensional vector spaces with bases $\{v_1, ..., v_n\}$ and $\{w_1, ..., w_m\}$ then there is a canonical isomorphism*

$$\begin{aligned} \operatorname{Bilin}_k(V, W) &\xrightarrow{\sim} \operatorname{Mat}_{n,m}(k) \\ \psi &\to \left(\psi(v_i, w_j)\right)_{ij} \end{aligned}$$

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*In particular a bilinear map $\psi$ is determined uniquely by the values $\psi(v_i, w_j)$.*

**Proposition 3.3.148 (Dual maps)**
*Let $V$ and $W$ be vector spaces, then there is a natural bijection*

$$\begin{array}{ccccc} \mathrm{Mor}_k(V, W^\vee) & \leftrightarrow & \mathrm{Bilin}_k(V, W) & \leftrightarrow & \mathrm{Mor}_k(W, V^\vee) \\ \psi_L & \leftarrow & \psi & & \\ & & \psi & & \to \psi_R \end{array}$$

*where*

$$\psi_L(v)(w) = \psi(v, w) = \psi_R(w)(v)$$

*When $V, W$ are finite-dimensional then $\psi_L$ is an isomorphism if and only if $\psi_R$ is an isomorphism. In this case we say $\psi$ is a perfect pairing. More generally*

$$\mathrm{rank}_k(\psi_L) = \mathrm{rank}_k(\psi_R)$$

*Proof.* The bijections stated are obvious. One may show that $\psi_L = \psi_R^\vee \circ \eta_V$ where $\eta_V : V \to V^{\vee\vee}$ is the dual isomorphism. Therefore $\psi_L$ is an isomorphism if and only if $\psi_R^\vee$ is an isomorphism, and by (3.3.145) if and only if $\psi_R$ is an isomorphism. Since $\eta_V$ is surjective we have $\mathrm{rank}_k(\psi_L) = \mathrm{rank}_k(\psi_R^\vee) = \mathrm{rank}_k(\psi_R)$, by (3.3.144). ∎

**Proposition 3.3.149**
*Let $\psi : V \times V \to k$ be a bilinear map. Then the following are equivalent*

*a) $\psi$ is a perfect pairing*
*b) $\psi_L$ is an isomorphism*
*c) $\psi_R$ is an isomorphism*
*d) $\psi_L$ is injective*
*e) $\psi_R$ is injective*
*f) $\psi_L$ is surjective*
*g) $\psi_R$ is surjective.*

*Proof.* We may use the fact $\dim_k V = \dim_k V^\vee$ and (3.3.136) to show b) ⟺ d) ⟺ f) and c) ⟺ e) ⟺ g). We've already shown that a) ⟺ b) ⟺ c). ∎

**Definition 3.3.150 (Orthogonal Complement)**
*Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces. Suppose $U \subset V$ is a subspace then define the* ***orthogonal complement***

$$U^\perp := \{w \in W \mid \psi(v, w) = 0 \quad \forall v \in U\}$$

**Proposition 3.3.151**
*Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces and $U \subset V$ a subspace. Then*

$$\dim_k U + \dim_k U^\perp = \dim_k V$$

*Indeed $\psi_R$ induces an isomorphism $U^\perp \to U^\circ$.*

*Proof.* We claim that $\psi_R(U^\perp) = U^\circ$. For if $w \in U^\perp$ then $\psi_R(w)(v) = \psi(w, v) = 0$ for all $v \in U$, and so $\psi_R(w) \in U^\circ$. Conversely given $\theta \in U^\circ$, as $\psi_R$ is surjective, there is $w \in W$ such that $\psi_R(w) = \theta$. By definition $w \in U^\perp$ as required.

As $\psi_R$ is injective then $\dim_k U^\perp \overset{(3.3.130)}{=} \dim_k U^\circ \overset{(3.3.143)}{=} \dim_k V/U = \dim_k V - \dim_k U$. ∎

**Remark 3.3.152**
*In the case $V = W$, then it's not necessarily true that $U \cap U^\perp = \{0\}$, and so $U^\perp$ is not necessarily a complementary subspace.*

*The classic example is the perfect pairing on $\mathbb{R}^n$ induced by $vDv^T$ for a real diagonal matrix $D$. Then it's true in general if and only if $D$ is positive-definite.*

**Proposition 3.3.153 (Quotients are dual to subspaces)**
*Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces. Suppose $U \subset V$ is a subspace, then there is a canonical perfect pairing*

$$\psi' : V/U \times U^\perp \to k$$

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*given by*

$$\psi'(v+U,w)=\psi(v,w)$$

*Proof.* The given map is well defined, for suppose $v_1+U=v_2+U$ then $v_1-v_2\in U\implies\psi(v_1-v_2,w)=0\quad\forall w\in U^\perp\implies\psi(v_1,w)=\psi(v_2,w)$ as required. It's clearly $k$-bilinear.

It's clear that $\psi'_R$ is injective, because $\psi'_R(w)=0_{V/U}\implies\psi_R(w)=0_V\implies w=0$.

By the previous Proposition $\dim_k U^\perp=\dim_k V/U$. Therefore by (3.3.136) $\psi'_R$ is an isomorphism and $\psi'$ is perfect.∎

## 3.3.17 Inverse Limits

We record the basic construction of an inverse limit of a tower of rings, modules, or algebras, as a concrete model for the categorical inverse limit (2.6.129), together with its functoriality and its compatibility with products; these are used repeatedly to build completions later on.

**Definition 3.3.154 (Inverse Limit of a Tower)**
*Let $\mathcal{C}$ be one of* **Ring**, *$A$-***Mod** *(for a fixed ring $A$), or $\mathrm{Alg}_A$ (for a fixed ring $A$, so in particular* $\mathbf{Alg}_k$*) – each an algebraic category (2.6.184) (via the usual axioms; (2.6.185), (2.6.186), (2.6.187) respectively). Let $\cdots\twoheadrightarrow M_2\twoheadrightarrow M_1\twoheadrightarrow M_0$ be a sequence of morphisms $\rho_n:M_n\to M_{n-1}$ of $\mathcal{C}$ $(n\geq 1)$. Define*

$$\varprojlim_n M_n:=\left\{(m_n)_{n\geq0}\in\prod_{n\geq0}M_n\;\middle|\;\rho_n(m_n)=m_{n-1}\text{ for every }n\geq1\right\},$$

*a sub-object of $\prod_n M_n$ in $\mathcal{C}$ (a subring, submodule, or subalgebra respectively, the ambient operations acting componentwise), with the evident projections $\lambda_n:\varprojlim_m M_m\to M_n$. Writing $\rho_{n,n}:=1_{M_n}$ and $\rho_{n,m}:=\rho_{m+1}\circ\cdots\circ\rho_n:M_n\to M_m$ for $n>m$ (composites of consecutive $\rho_k$'s), every $(m_n)_{n\geq0}\in\varprojlim_n M_n$ satisfies the* ***repeated compatibility***

$$\rho_{n,m}(m_n)=m_m\quad\text{for all }n\geq m,$$

*not just for consecutive $n,m$: by induction on $n-m$, composing the defining relations $\rho_k(m_k)=m_{k-1}$ for $m<k\leq n$.*

**Lemma 3.3.155 (This Realises the Categorical Inverse Limit)**
*In the setting of (3.3.154), regard $I:=\mathbb{N}$ (usual order) as a directed preorder and the tower as an inverse system $F:I^{op}\to\mathcal{C}$, $F(n):=M_n$, with restriction maps $\rho_{n,m}$ as in (3.3.154). Then $\left(\varprojlim_n M_n,\{\lambda_n\}\right)$ is an inverse limit for $F$ in the sense of (2.6.129).*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

$\mathcal{C}$ is an algebraic category (2.6.184), in each of the three cases (as recalled in (3.3.154)). By (2.6.191), $\varprojlim_n F(n)$ exists in $\mathcal{C}$, with underlying set $\left\{(m_n)_n\in\prod_n M_n:\rho_{n,m}(m_n)=m_m\text{ for all }n\geq m\right\}$ and the ambient operations acting coordinatewise; by the repeated compatibility (3.3.154), this defining condition holds for all $n\geq m$ as soon as it holds for consecutive $n=m+1$, so this is exactly the sub-object $\varprojlim_n M_n$ of (3.3.154), with the same (componentwise) structure and projections. So $\left(\varprojlim_n M_n,\{\lambda_n\}\right)$ is this categorical inverse limit. ∎

**Corollary 3.3.156 (Functoriality of the Inverse Limit)**
*Let $\mathcal{C}$ be as in (3.3.154), $(A_n,\rho_n)_{n\geq0}$ and $(B_n,\tau_n)_{n\geq0}$ towers in $\mathcal{C}$, and for each $n\geq0$ let $\sigma_n:A_n\to B_n$ be a morphism of $\mathcal{C}$ such that, for every $n\geq1$, the square*

$$\begin{array}{ccc} A_n & \xrightarrow{\sigma_n} & B_n \\ {\scriptstyle\rho_n}\downarrow & & \downarrow{\scriptstyle\tau_n} \\ A_{n-1} & \xrightarrow[\sigma_{n-1}]{} & B_{n-1} \end{array}$$

*commutes, i.e. $\tau_n\circ\sigma_n=\sigma_{n-1}\circ\rho_n$. Then*

$$\varprojlim\sigma:\varprojlim_n A_n\longrightarrow\varprojlim_n B_n,\quad(a_n)_n\mapsto(\sigma_n(a_n))_n$$

*is a well-defined morphism of $\mathcal{C}$; and if every $\sigma_n$ is an isomorphism, so is $\lim_{\leftarrow}\sigma$.*

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*Proof.* By (3.3.155), $\lim_{\leftarrow n} A_n$ and $\lim_{\leftarrow n} B_n$ (with their projections) are inverse limits, in the sense of (2.6.129), of the inverse systems $F, G : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ built from the two towers; the commuting squares above are exactly the naturality hypothesis of (2.6.132) for the family $\sigma = (\sigma_n)_n$ (composing squares for consecutive $n$ gives it for the composite restriction maps $\rho_{n,m}$, $\tau_{n,m}$, $n > m$, as needed there). That corollary gives the induced morphism $\lim_{\leftarrow n} \sigma_n$, agreeing on each coordinate with $\sigma_n$ by construction of the induced map on limits-as-sets (2.6.130) – that is, the map $(a_n)_n \to (\sigma_n(a_n))_n$ above – and an isomorphism whenever every $\sigma_n$ is. ∎

**Proposition 3.3.157 (Inverse Limits Commute with Products)**
*Let $\mathcal{C}$ be as in (3.3.154), $K$ a set, and for each $k \in K$ a tower $(M_{k,n}, \rho_{k,n})_{n\geq 0}$ in $\mathcal{C}$. Then $\left(\prod_{k\in K} M_{k,n}\right)_{n\geq 0}$, with restriction maps $\prod_{k\in K} \rho_{k,n} : \prod_k M_{k,n} \to \prod_k M_{k,n-1}$, is again a tower in $\mathcal{C}$, and*

$$\Phi : \prod_{k\in K} \lim_{\leftarrow n} M_{k,n} \longrightarrow \lim_{\leftarrow n} \left(\prod_{k\in K} M_{k,n}\right), \quad \Phi\big((x_k)_{k\in K}\big) := \Big(\big(x_{k,n}\big)_{k\in K}\Big)_{n\geq 0} \quad \big(x_k = (x_{k,n})_{n\geq 0}\big)$$

*is a well-defined isomorphism of $\mathcal{C}$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Write $F_k : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ for the inverse system of the tower $(M_{k,n})_n$, so $\left(\lim_{\leftarrow n} M_{k,n}, \{\lambda_{k,n}\}_n\right)$ realises $\lim_{\leftarrow} F_k$ (3.3.155), and likewise $\left(\lim_{\leftarrow n}\left(\prod_k M_{k,n}\right), \{\Lambda_n\}_n\right)$ realises $\lim_{\leftarrow}\left(\prod_k F_k\right)$, where $\Lambda_n$ is projection onto the $n$-th coordinate. Since $\prod_{k\in K} \lim_{\leftarrow n} M_{k,n}$ (with the evident coordinate projections $p_k$) is the categorical product of the $\lim_{\leftarrow n} M_{k,n}$ in $\mathcal{C}$, (2.6.133) gives that $\lim_{\leftarrow n}\left(\prod_k M_{k,n}\right)$ exists (as it already does, here) and that there is a *unique morphism* $\Phi'$ (of $\mathcal{C}$, not assumed a priori to be an isomorphism) such that

$$\pi_{k,n} \circ \Lambda_n \circ \Phi' = \lambda_{k,n} \circ p_k \quad \text{for all } k \in K, n \geq 0,$$

where $\pi_{k,n} : \prod_{k'} M_{k',n} \to M_{k,n}$ is the product projection; and that this $\Phi'$ is an isomorphism.

The explicit $\Phi$ above satisfies exactly this defining property: for $(x_k)_k \in \prod_k \lim_{\leftarrow n} M_{k,n}$ (so $x_k = (x_{k,n})_n$), both $\pi_{k,n}\big(\Lambda_n\big(\Phi\big((x_k)_k\big)\big)\big)$ and $\lambda_{k,n}\big(p_k\big((x_k)_k\big)\big)$ equal $x_{k,n}$. By uniqueness of the morphism with this property, $\Phi = \Phi'$; as $\Phi'$ is an isomorphism, so is $\Phi$. ∎

**Corollary 3.3.158 (Cofinal Reindexing of a Tower)**
*Let $\mathcal{C}$ be as in (3.3.154), $(M_n, \rho_n)_{n\geq 0}$ a tower in $\mathcal{C}$, and $n_0 < n_1 < n_2 < \cdots$ a strictly increasing sequence in $\mathbb{N}$. Then, writing $\tilde{\rho}_N := \rho_{n_{N+1},n_N} : M_{n_{N+1}} \to M_{n_N}$ for the composite restriction map of (3.3.155), $\left(M_{n_N}, \tilde{\rho}_N\right)_{N\geq 0}$ is again a tower in $\mathcal{C}$, and*

$$(m_n)_{n\geq 0} \mapsto \left(m_{n_N}\right)_{N\geq 0}$$

*defines an isomorphism $\lim_{\leftarrow n} M_n \xrightarrow{\sim} \lim_{\leftarrow N} M_{n_N}$ of $\mathcal{C}$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Regard $I := \mathbb{N}$ as a directed preorder and, per (3.3.155), the tower $(M_n)_n$ as an inverse system $F : I^{\mathrm{op}} \to \mathcal{C}$ realising $\lim_{\leftarrow n} M_n$. Let $I' := \{n_N : N \geq 0\} \subset I$, a full subcategory: it is directed (a subset of the totally ordered set $I$), and cofinal, as $n_N \to \infty$ so every $n \in I$ satisfies $n \leq n_N$ for some $N$ (2.6.126). The order-preserving bijection $\sigma : \mathbb{N} \to I'$, $\sigma(N) := n_N$, is an isomorphism of the directed preorders $\mathbb{N}$ and $I'$, and $F\sigma : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ is exactly the tower $\left(M_{n_N}, \tilde{\rho}_N\right)_N$ (its restriction map for $N < N+1$ is $F(\sigma(N) \to \sigma(N+1)) = \rho_{n_{N+1},n_N} = \tilde{\rho}_N$, and composites thereof match for $N < N'$); so, by (2.6.98) (applied to $\sigma$) together with (3.3.155) applied to the tower $\left(M_{n_N}, \tilde{\rho}_N\right)_N$, $\left(\lim_{\leftarrow n} M_n, \{\lambda_{n_N}\}_{N\geq 0}\right)$ realises $\lim_{\leftarrow i\in I'} F(i)$, and $\left(\lim_{\leftarrow N} M_{n_N}, \{\lambda'_N\}_{N\geq 0}\right)$ (the latter's own projections) also realises it; identifying these two cones over $F|_{I'^{\mathrm{op}}}$ via the (unique, by terminality) comparison isomorphism between them, which sends $(m_n)_{n\geq 0} \in \lim_{\leftarrow n} M_n$ (viewed as a cone-point of the first) to the compatible sequence $\left(m_{n_N}\right)_{N\geq 0} \in \lim_{\leftarrow N} M_{n_N}$ agreeing with it on projections $\lambda_{n_N} = \lambda'_N \circ$ (this isomorphism).

By (2.6.131) (as $I'$ is cofinal in $I$), the canonical morphism of cones $\lim_{\leftarrow i\in I} F(i) \to \lim_{\leftarrow i\in I'} F(i)$ is an isomorphism, and by construction (restriction of the limit cone's own projections $\lambda_n$ to $i = n_N \in I'$) it sends $(m_n)_{n\geq 0} \to \left(m_{n_N}\right)_{N\geq 0}$ under the identification of the previous paragraph. Composing the two isomorphisms of this and the previous paragraph gives the stated isomorphism $\lim_{\leftarrow n} M_n \xrightarrow{\sim} \lim_{\leftarrow N} M_{n_N}$, $(m_n)_n \to \left(m_{n_N}\right)_N$. ∎

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